Universidad Politécnica de Madrid

Escuela Superior de Ingeniería de Sistemas Informáticos

Solving the Multimodal Capacitated Vehicle Routing Problem (CVRP)

Subject: Bioinspired Algorithms for Optimization (BAO)

Group ID: 9

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Contents

1 Abstract

This project addresses the Multimodal Capacitated Vehicle Routing Problem (CVRP) as specified for Group 9. The objective is to design and implement bioinspired metaheuristics—specifically Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Genetic Algorithms (GA)—to provide multiple high-quality and structurally diverse routing alternatives. Our methodology involves rigorous hyperparameter tuning, localized 2-opt memetic refinement, and multi-problem statistical benchmarking across 35 independent executions to evaluate algorithmic robustness.

We engineered and evaluated distinct constraint-handling and niching strategies: geometric clustering for the PSO to reduce dimensionality, passive archive filtering with strict feasibility bounds for the ACO, and deterministic crowding for the GA. The final global analysis across the dataset reveals a distinct performance division. The **Archive ACO** stands as the unequivocally superior optimizer for routing quality and stability, successfully leveraging its native graph-traversal logic to minimize logistical costs. Conversely, the **Complete Clustering PSO** proved to be the most effective generator of extreme structural diversity (Jaccard $> 90\%$). The Memetic GA, while computationally the fastest by a significant margin, struggled with the rigid geometric constraints of larger CVRP instances, resulting in high variance. Ultimately, the empirical results demonstrate that strict capacity feasibility rules paired with passive memory archiving offer the most reliable mathematical framework for solving multimodal logistics networks.

2 Problem

2.1 Background

The Capacitated Vehicle Routing Problem (CVRP) is a cornerstone of operations research and supply chain logistics. It involves designing an optimal set of delivery routes from a central depot to a set of geographically dispersed customers, ensuring that the total accumulated demand of any route does not exceed the maximum capacity C of the assigned vehicle. Minimizing the total distance traveled directly translates to reduced fuel consumption, lower operational costs, and minimized environmental impact.

Mathematically, the CVRP is a well-known NP-Hard combinatorial optimization problem. As the number of customers grows, the search space expands factorially. Traditional exact optimization methods become computationally intractable for large-scale scenarios (such as the modern Uchoa et al. instances containing up to 916 nodes used in this study). Consequently, the implementation of advanced bioinspired metaheuristics—such as Particle Swarm Optimization, Ant Colony Optimization, and Genetic Algorithms—is essential to discover near-optimal solutions within reasonable execution times.

Beyond the standard CVRP, this research specifically addresses the **Multimodal CVRP**. In real-world logistics, returning a single mathematical "optimum" is often impractical; unforeseen disruptions like traffic congestion, severe weather, or road closures can render the primary fleet plan invalid. Therefore, the objective shifts from finding a singular optimum to discovering a robust set of high-quality, structurally diverse alternatives. Specifically, the goal of this project is to extract at least three distinct routing plans that maintain a total distance within 5% of the best-found solution, whilst exhibiting significant topological variations in their customer-to-vehicle assignments.

2.2 Problem Definition

In this assignment, we solve the Multimodal CVRP (Group 9 variant). The goal is to produce three distinct, high-quality solutions for each instance. A solution is valid if every customer is visited exactly once, all routes start and end at the depot (0,0), and no vehicle exceeds its capacity. The multimodal requirement dictates that:

- All three alternatives must have a total distance within 5% of the best-found solution.
- Alternatives must be structurally diverse, meaning customer-to-route assignments must differ significantly.

3 Algorithm Design

3.1 Metaheuristics Used

To solve the Multimodal CVRP, we implemented three distinct bioinspired metaheuristics: Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and a Genetic Algorithm (GA).

Particle Swarm Optimization (PSO): Our research focuses on adapting PSO to combinatorial spaces. While traditionally continuous, PSO's swarm dynamics provide a unique approach to route exploration. We developed a multi-stage architecture to overcome dimensionality issues and meet multimodal requirements.

Ant Colony Optimization (ACO): Unlike PSO, which maps a continuous space to a discrete problem, Ant Colony Optimization is natively designed for graph-based routing. It constructs solutions constructively, node-by-node, using simulated pheromone trails and heuristic visibility. We implemented this metaheuristic because its probabilistic graph-traversal nature aligns perfectly with the CVRP. To meet Group 9's requirements, we extended the standard architecture with specialized multimodal memories and sequential restarts to prevent the colony from converging on a single, dominant trail.

Genetic Algorithm (GA): Unlike the velocity-based continuous transformations of PSO or the probabilistic, edge-by-edge construction of ACO, the Genetic Algorithm evaluates and recombines entire solution topologies simultaneously. We implemented this metaheuristic because its evolutionary "building block" hypothesis, where crossover operators preserve highly efficient customer sub-routes while mutation explores new sequences, is exceptionally well-suited for discrete permutation spaces. To satisfy the multimodal requirements, our GA architecture was heavily upgraded into a Memetic Algorithm (integrating localized 2-opt repair) and reinforced with explicit niching mechanisms (such as deterministic crowding and sequential tabu memory) to force the co-evolution of structurally distinct logistical plans.

3.2 Codification Used

The codification strategies differ significantly across the algorithms:

- **PSO Representation:** A particle's position is defined as a permutation of customers (a "Giant Tour"). The movement logic is adapted to this discrete space by redefining "Velocity" as a sequence of transposition operators (swaps).
  - **Probabilistic Movement:** To move a particle, we calculate the swap sequences required to reach *pBest* and *gBest*. Unlike the continuous version, the parameters w, c_1, and c_2 are interpreted as **probabilities (range** $[0, 1]$**)**. These probabilities dictate whether a specific swap from the sequence is actually performed, allowing for a stochastic balance between momentum and attraction to optima.
  - **Constraint Handling (Capacity Limits):** Managing the vehicle capacity constraint depends on the algorithmic phase:

- * *Standard Decoding:* For full-map approaches or within isolated clusters, the permutation is decoded using a **splitting procedure**. It iterates through the tour and inserts a return to the depot whenever adding the next customer would exceed the maximum vehicle capacity C.
  * *Clustering Consolidation:* In Phase 2, independent feasible routes from each cluster are merged using the `JoinRoutes` mechanism to reduce the total fleet size. However, as noted in our analysis, this heuristic occasionally struggles to merge boundary routes without slightly violating capacity constraints if the initial K-Means partitions were not perfectly balanced, highlighting a known limitation of structural partitioning.

- **ACO Representation & State Transition:** A solution is constructed directly as a sequence of visited nodes. An ant selects the next customer j from its current position i probabilistically, proportional to $[\tau_{ij}]^\alpha[\eta_{ij}]^\beta$, where τ_{ij} is the pheromone level and $\eta_{ij} = 1/d_{ij}$ is the heuristic visibility.
  - **Constraint Handling (Feasibility vs. Penalty):** To manage the vehicle capacity constraint C, the ACO employs two distinct evaluated strategies:
    * *Feasibility (Dynamic):* The feasible neighborhood is strictly filtered during construction. An ant can only transition to customers whose demand fits the remaining truck capacity; otherwise, it is forced to return to the depot (node 0) to reset its load.
    * *Penalty (Soft Constraint):* Ants construct routes freely, but upon completion, a massive mathematical penalty is applied to the objective function if the total route violates fleet or capacity limits. This forces the colony to learn the boundaries through pheromone evolution rather than hard-coded rules.

- **GA Representation & Genetic Operators:** The GA operates natively on the discrete combinatorial space.
  - **Chromosome:** A solution is codified as a pure permutation of all customer IDs (a "Giant Tour").
  - **Recombination (OX1):** Order Crossover is used to inherit structural sub-routes from parents, ensuring no customers are duplicated or omitted.
  - **Mutation (Inversion):** Sub-sequences of the tour are reversed. This acts as a high-energy disruption to untangle crossed edges and escape local optima.
  - **Constraint Handling:** The Giant Tour is decoded into valid routes using an **Greedy Splitter** approach. The decoder traverses the chromosome and automatically inserts a return to the depot whenever adding the next customer would exceed the maximum vehicle capacity C, guaranteeing 100% mathematically feasible solutions.

3.3 Implementation Details

Particle Swarm Optimization (PSO): The PSO was implemented following a two-phase strategy. Phase 1 tested full-map exploration using multimodal techniques (*Clearing*, *Fitness Sharing*, *Species Competition*, and *Sequential Niching*).

To address the high dimensionality of the CVRP, Phase 2 introduced a **Clustering-based architecture** to combat the capacity constraint by "dividing and conquering":

1. **K-Means Clustering Engine:** We use K-Means to partition the map into k subproblems. Three specialized strategies were developed:

- **Normal Clustering:** Groups nodes based on Euclidean distance.
- **Angular Clustering:** Uses a sweep-line approach based on polar coordinates (angles) relative to the depot.
- **Complete Clustering:** A 5-dimensional approach that balances spatial distribution and vehicle capacity to create highly solvable sub-problems.

2. **Localized Search:** Each cluster is solved independently by a PSO with *Sequential Niching*. This ensures that the search space is small enough for the PSO to converge efficiently while extracting the required multimodal alternatives.

3. **Route Joining:** A `JoinRoutes` utility consolidates the independent results into a final global fleet plan.

Ant Colony Optimization (ACO): The base engine was developed following a standard Ant System (AS) architecture. However, because standard ACO naturally converges towards a single heavy pheromone trail, we implemented two distinct multimodal strategies. Combined with our two constraint handlers, this resulted in four testable configurations:

1. **Sequential Niching:** The colony explores and converges on a high-quality global route. After a set number of iterations, the pheromone matrix is cleared or heavily penalized (rebooted). This forces the ants to abandon the previous global best and discover entirely new routing topologies in subsequent cycles.

2. **Archive Strategy:** The colony runs continuously without resetting pheromones, relying instead on an external "Archive" to filter and save diverse solutions. Whenever a high-quality route is generated, its structural diversity is calculated via Jaccard distance against the already archived routes. It is only accepted if it provides a genuinely new topological alternative.

This yields four algorithmic variants: *Feasibility + Archive*, *Feasibility + Sequential*, *Penalty + Archive*, and *Penalty + Sequential*. Unlike the PSO, which applies local search only to top candidates, the ACO acts as a memetic hybrid: **2-opt local search** is applied to every route constructed by every ant during the generation phase. This ensures that pheromones are only deposited on geometrically optimized paths, drastically accelerating convergence.

Genetic Algorithm (GA): The base engine was implemented as a *Memetic Algorithm*. To enforce the extraction of three structurally diverse niches, we evaluated two distinct population-control architectures:

1. **Deterministic crowding:** A passive, single-population strategy. During the replacement phase, a child only replaces its most structurally similar parent (measured via Jaccard distance) if it has a better fitness. This creates localized "niche competition," allowing multiple distinct routing topologies to safely co-exist in the same gene pool.

2. **Sequential tabu niching:** An active, multi-interval strategy. Once the GA converges on an optimum, its topology is saved to an external Tabu List. In subsequent execution intervals, any chromosome structurally similar to these Tabu niches receives a massive fitness penalty, actively forcing the population to explore entirely new geographical zones.

Furthermore, both configurations heavily rely on **2-opt local search**. As demonstrated in our experiments, this memetic operator is strictly essential when using *Soft Constraints (Penalties)*, as it successfully "heals" severely overloaded, invalid chromosomes back into 100% capacity-compliant routes over successive generations.

Multimodal Validation Logic (Common for all): Final validation is performed externally in the analysis notebook. The three selected alternatives must be within 5% of the global best distance and exhibit significant structural diversity, calculated via the **Jaccard distance** of the routes' edges.

4 Experiments Design

4.1 Parameter Settings

A systematic **Parameter Fine-Tuning** was conducted to optimize performance:

Particle Swarm Optimization (PSO):

- **Core Probabilities:** Tuning of w, c_1, c_2 focused on ensuring enough exploration in early stages and exploitation in late stages.
- **Species Competition (Phase 1):** Tuning of the species radius and penalty thresholds to prevent species overlap.
- **Complete Clustering (Phase 2):** Calibration of the number of clusters and the *Sequential Niching* radius to balance the complexity of each sub-problem with the quality of the joined routes.

Ant Colony Optimization (ACO):

- **Core Heuristics (α, β):** Tuning the balance between the pheromone trail influence (α) and the greedy distance-based visibility (β) to ensure routes are geometrically logical but historically informed.
- **Pheromone Dynamics (ρ, Q):** Calibration of the evaporation rate (ρ) and the deposit multiplier (Q) to control the colony's convergence speed and prevent premature stagnation on suboptimal local minima.
- **Multimodal Thresholds:** Fine-tuning the Jaccard distance minimum (`min_diversity`) and the maximum acceptable fitness degradation (`quality_threshold` ≤ 1.05) required to accept a route into the Archive, ensuring structural variety without violating the 5% constraint.

Genetic Algorithm (GA):

- **Exploitation vs. Exploration Balance (P_m, K):** Through sensitivity analysis, we determined that the *Exploitation Focus* ($P_m = 0.05$, $K = 5$) was superior for Sequential Niching. High selective pressure (K) and low mutation (P_m) are required to stabilize and optimize the sub-optimal regions forced by the Tabu memory erasure.
- **Memetic Intensity (P_{ls}):** Tuning the local search probability to 0.80. This high intensity is critical for the repairing mechanism, ensuring that chromosomes exploring forbidden overloaded paths (soft constraints) are geometrically repaired into feasible, capacity-compliant routes.
- **Tabu/Jaccard Thresholds:** Calibration of the structural diversity barrier. We set a minimum Jaccard distance to prevent the population from re-converging on previously extracted optima, balancing the trade-off between strict topological uniqueness and acceptable fitness degradation.

4.2 Experimental Setup

1. **Execution Protocol:** Every instance was executed **35 independent times** to account for algorithmic stochasticity. For the final comparison, we selected the *Baseline PSO*, the *Species Competition PSO*, and the *Complete Clustering PSO* alongside the *Archive* and *Sequential* ACO variants.
2. **Data Logging (CSV Metrics):** The top 3 alternatives are refined via **2-opt local search** to eliminate edge crossings. The logged metrics are:

- `mean_fitness` / `std_fitness`: Average distance and stability of the alternatives.
- `mean_jaccard_distance`: Quantification of structural variety.
- `n_evaluations` / `n_generations`: Computational effort (cumulative for clustering).
- `tiempo`: Total sequential runtime in seconds.

3. **Statistical Analysis:** We employed the **Friedman aligned ranks test** for global significance across multiple algorithms, and the **Shaffer post-hoc test** for pairwise comparisons to establish statistical superiority. When comparing strictly two variants, the **Wilcoxon signed-rank test** was utilized.

4. **Limitations of the Setup:** The experimental environment has notable limitations. First, tracking the exact Jaccard diversity continuously across all generations is computationally prohibitive for the ACO; thus, diversity is strictly evaluated on the final archived niches. Second, evaluating structural routing relies heavily on 2-opt post-processing; algorithms that naturally build crossing paths might be penalized in runtime, even if their underlying node sequence is mathematically promising. Finally, real-world CVRP execution is often parallelized, whereas our 'tiempo' metric strictly measures sequential execution speed.

4.3 Data Description

The experimentation utilized 100 benchmark instances from the Uchoa et al. "X" dataset, following the `X-n[T]-k[V].vrp` format. These instances range from 101 to 916 nodes, providing a comprehensive evaluation of the algorithms' scalability and their ability to handle varying customer demands and vehicle capacities while maintaining the required 5% quality threshold for multimodal outputs.

5 Experimental Results

This section details the performance analysis of the implemented metaheuristics. We first evaluate the internal variants of each algorithm independently to determine their best configuration, followed by a global comparative analysis of the three optimal approaches.

5.1 PSO Internal Comparison

To systematically evaluate the performance of the various PSO architectures, an incremental analysis was conducted, transitioning from a standard baseline configuration to more sophisticated hybridized models.

To validate these approaches, we utilize the **X-n106-k14** benchmark instance as a primary case study. This problem possesses a known optimal fitness of **26362**. Based on this ideal value, the multimodal quality threshold for this specific instance is established at **27680** (representing the maximum 5% deviation allowed by the Group 9 requirements). Any solution exceeding this value is considered suboptimal for the purposes of this study.

5.1.1 Phase 1: Baseline Evaluation and the Necessity of Local Search (2-opt)

The initial experiment consisted of a standard PSO featuring a Star Topology (Global Best) applied to the complete map of a smaller instance. The objective was to assess its raw routing capabilities and its natural inclination toward multimodal diversity.

As observed in the raw output maps, the unrefined PSO struggles significantly with spatial geometry. The generated routes (averaging a fitness of $\sim$ 301.77) exhibit severe "spaghetti" topologies, characterized by numerous self-intersecting edges. This occurs because the standard PSO explores permutations stochastically but lacks a geometric mechanism to untangle inefficient local crossings.

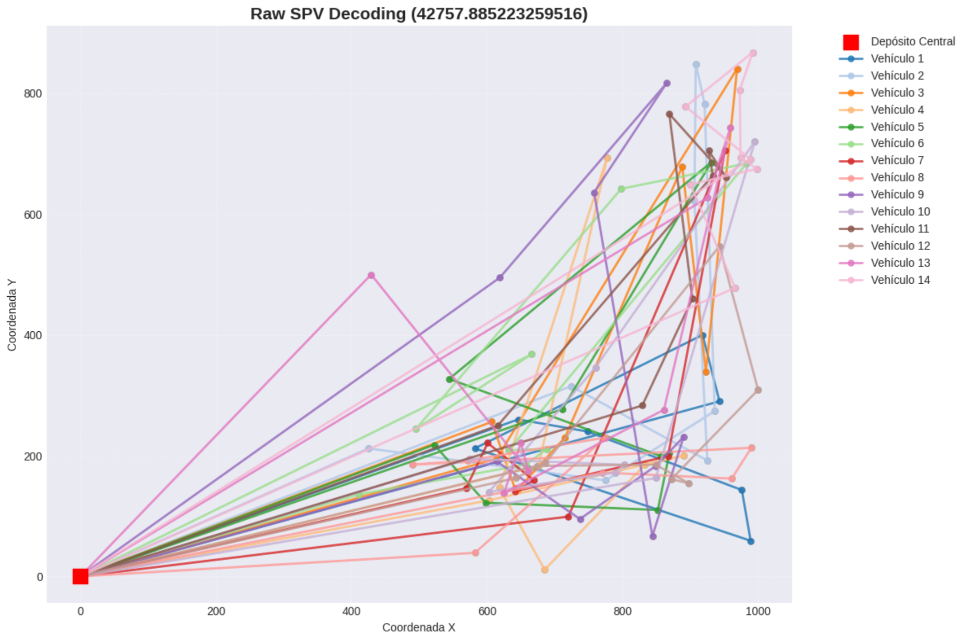


Figure 1: Showing a solution without 2-opt.

To mitigate this, a **2-opt local search heuristic** was systematically applied to the top candidates to refine edge crossings without altering the global cluster assignments.

The application of 2-opt drastically improved the geometric efficiency of the routes, reducing the fitness and eliminating the chaotic intersections. This establishes that PSO alone is insufficient for the CVRP without local search refinement.

However, the primary failure of the Baseline PSO lies in its inability to satisfy the multimodal requirement. The structural diversity analysis of the refined routes revealed the following Jaccard distances:

- Jaccard Distance (Route 1 vs Route 2): **0.00%**
- Jaccard Distance (Route 1 vs Route 3): **92.31%**

A Jaccard distance of 0.00% indicates that Route 1 and Route 2 are topological clones. The swarm suffered from premature convergence—evident in the rapid flatlining of the convergence curve and the steady decline in swarm diversity—resulting in the algorithm effectively discovering only two unique topologies instead of the required three. Furthermore, the high variance observed in the final population's boxplot indicates that a massive portion of the swarm became hopelessly trapped in low-quality local optima.

This structural failure dictates the absolute necessity of implementing explicit multimodal techniques to force the swarm to discover multiple, non-overlapping optima.

5.1.2 Evaluation of Multimodal Techniques

Having demonstrated the structural failure of the standard Baseline, explicit multimodal techniques were introduced to the swarm to enforce diversity: *Clearing*, *Fitness Sharing*, *Species Competition*, and *Sequential Niching*.

Figure 5 visualizes the Top 3 alternative routes generated by each technique (after 2-opt refinement), allowing for a visual comparison of their multimodal routing capabilities on the full map.

Fitness Analysis vs. The Optimal Benchmark: As established, the acceptable 5% tolerance threshold for this instance is **27680**. Analyzing the outputs, none of the full-map

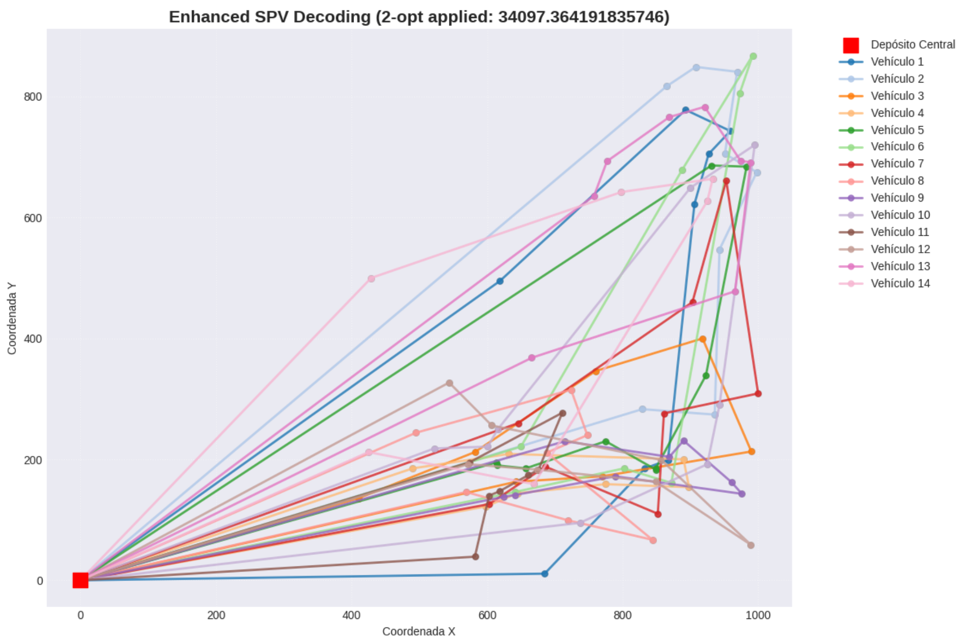


Figure 2: Showing a solution with 2-opt.

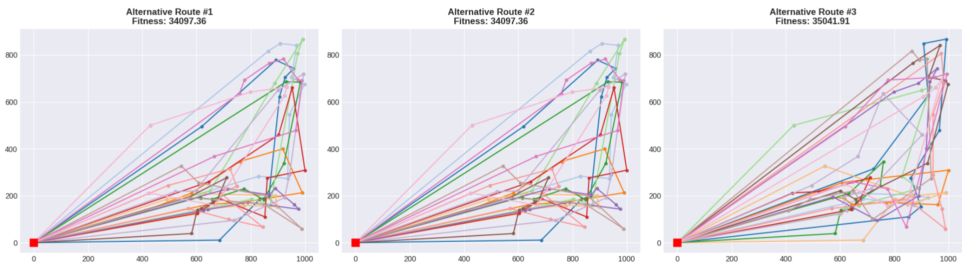
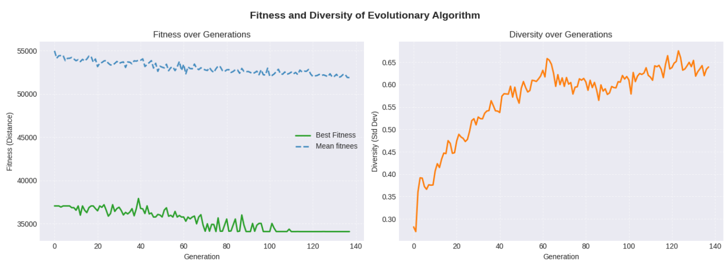
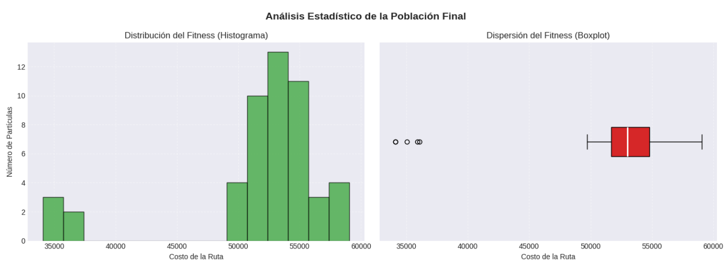


Figure 3: Showing best 3 solutions of baseline PSO.

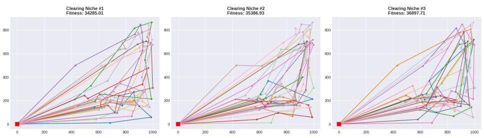


(a) Fitness and Diversity evolution (baseline)

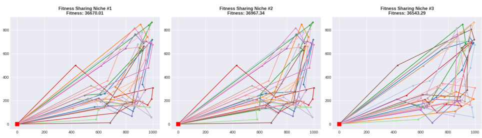


(b) Final population Histogram and Boxplot (baseline)

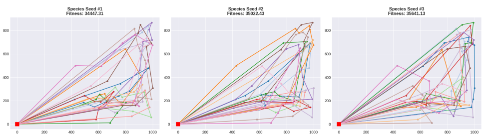
Figure 4: Evolution of PSO over generations (baseline).



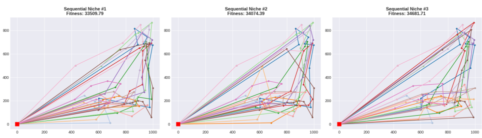
(a) Clearing Alternatives



(b) Fitness Sharing Alternatives



(c) Species Competition Alternatives

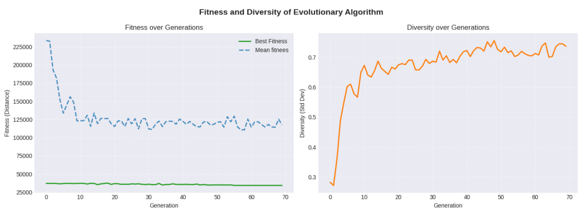


(d) Sequential Niching Alternatives

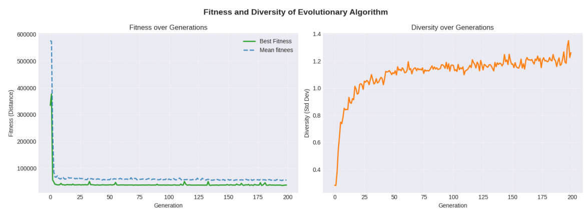
Figure 5: Comparison of the Top 3 structurally diverse alternatives extracted by each multimodal technique.

multimodal techniques successfully met this requirement. Sequential Niching performed the best relatively (ranging from 33509.79 to 34681.71), while Fitness Sharing and Clearing produced highly degraded routes (often exceeding 36000). The geometric inefficiency observed in these results stems directly from the intense penalization strategy applied (2× sum of distances from the depot). While this massive penalty successfully prevents the swarm from converging on a single local optimum, it introduces chaotic perturbations that prevent the particles from fine-tuning their routes.

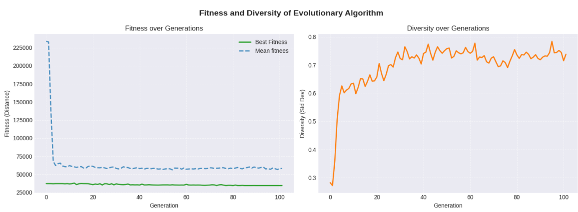
Figure 6 illustrates how this penalization affects the swarm's internal dynamics.



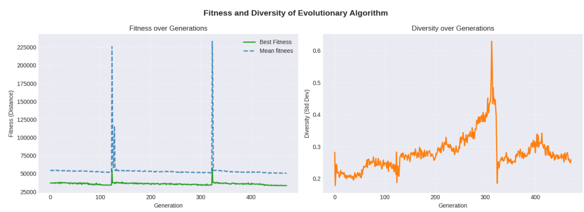
(a) Clearing



(b) Fitness Sharing



(c) Species Competition



(d) Sequential Niching

Figure 6: Evolution of Fitness and Diversity across the multimodal techniques.

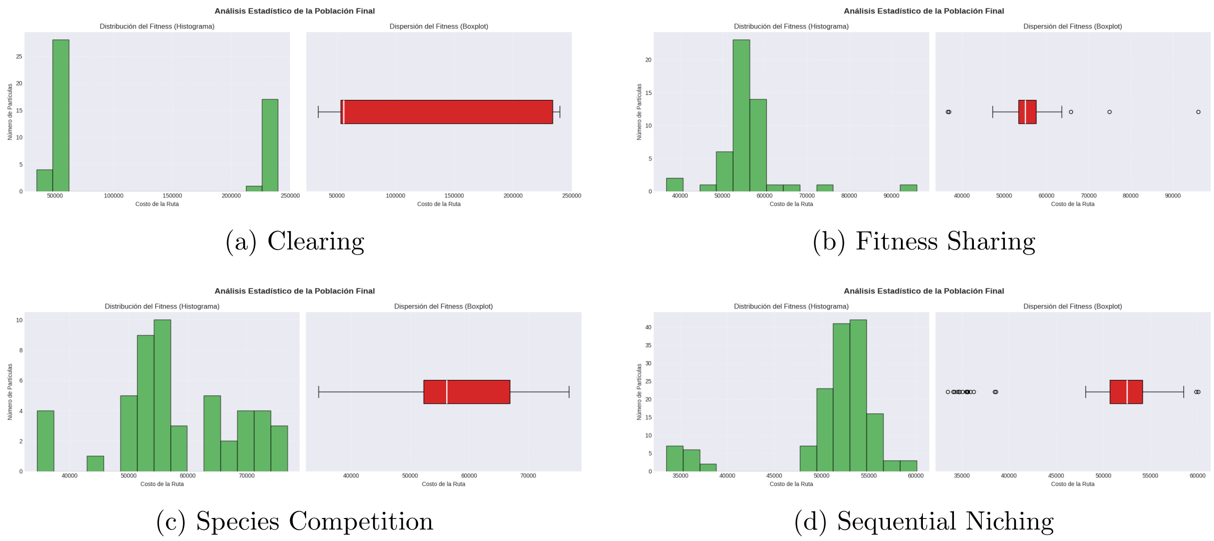
Diversity and Structural Analysis: In contrast to the Baseline's premature convergence, the multimodal curves demonstrate a massive success in maintaining swarm diversity. For instance, in both Clearing and Fitness Sharing, the diversity index shoots up early and remains highly elevated throughout all generations.

To objectively quantify this multimodal success, we calculated the Jaccard distances between the Top 3 niches of each algorithm:

- **Sequential Niching Jaccard:** Niche 1 vs 2 (56.63%), Niche 1 vs 3 (69.23%), Niche 2 vs 3 (72.04%).
- **Fitness Sharing Jaccard:** Niche 1 vs 2 (28.78%), Niche 1 vs 3 (90.32%), Niche 2 vs 3 (89.81%).

- **Clearing Jaccard:** Niche 1 vs 2 (90.83%), Niche 1 vs 3 (92.31%), Niche 2 vs 3 (89.30%).
- **Species Competition Jaccard:** Species 1 vs 2 (**93.27%**), Species 1 vs 3 (**92.31%**), Species 2 vs 3 (**91.82%**).

Figure 7 illustrates the fitness distribution of the final population.



(a) Clearing

(b) Fitness Sharing

(c) Species Competition

(d) Sequential Niching

Figure 7: Histogram and Boxplot of last population.

Phase 1 Conclusion: The multimodal techniques successfully solved the Baseline's diversity collapse, achieving highly distinct alternative topologies (Jaccard $> 50\%$). However, this structural diversity came at an unacceptable cost to routing quality, failing the 5% gap requirement. Enforcing multimodality on a massive, highly constrained global CVRP map forces the PSO to explore inefficient permutations to escape heavy penalties. This paradox definitively concludes Phase 1: a single global swarm cannot simultaneously satisfy strict capacity constraints, near-optimal fitness, and structural diversity. This dictates the transition to Phase 2, where **Clustering** must be used to partition the map into smaller, isolated spaces where multimodal algorithms can operate effectively without chaotic global penalties.

5.1.3 Phase 2: Clustering Architecture and the "JoinRoutes" Consolidation

Despite the success of the continuous multimodal techniques in generating diverse structural routes, their absolute fitness plateaued around the 33,000–36,000 range, significantly above the 27680 threshold. This gap is a known limitation of applying pure continuous algorithms via SPV (Smallest Position Value) mapping to the discrete CVRP. The standard PSO frequently struggles with spatial customer assignment, often placing geographically distant nodes into the same vehicle route, creating inefficient "spaghetti" topologies.

To overcome this, we introduce a **Cluster-first, Route-second** heuristic paradigm. Instead of allowing a single swarm to blindly map a high-dimensional continuous space into a massive discrete route, the geographical map is partitioned into smaller, logical sub-problems. An independent PSO swarm is then deployed exclusively within each isolated sub-problem. This drastically reduces the dimensional complexity and physically prevents the algorithm from assigning distant customers to the same vehicle.

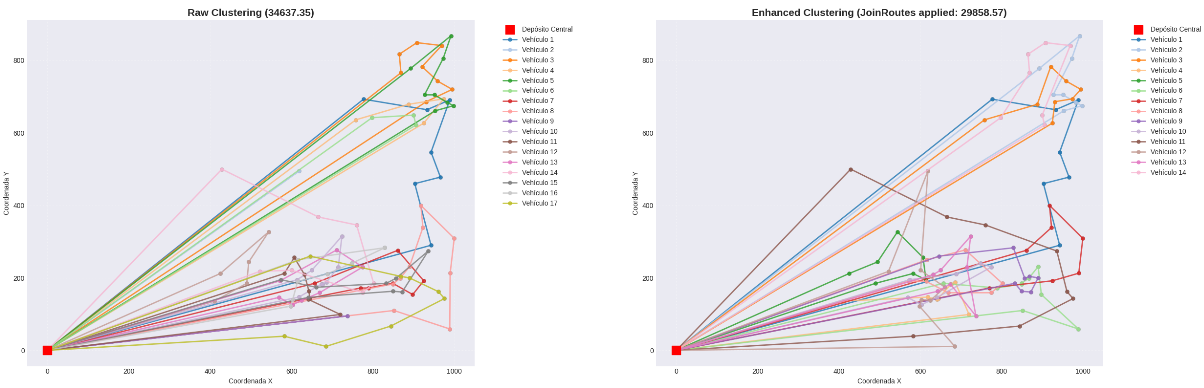
The Logic Behind the Truck Penalty: By dividing the global map into independent sub-problems, we inherently break the global fleet constraint. To prevent individual sub-swarms from taking the "easy route" by deploying extra, underutilized vehicles, we enforce a strict mathematical boundary:

$$Excess = \max(0, \text{Trucks Used} - \text{Truck Limit})^{\text{Truck Penalty}}$$

For each isolated cluster, the `ClusteringManager` calculates the absolute minimum number of vehicles required based on its total demand. If the localized PSO swarm fails to pack the vehicles efficiently and exceeds this theoretical limit, a massive quadratic penalty is applied. This acts as a mathematical whip, driving the swarm to pack the vehicles as densely as physically possible to approach the theoretical minimum.

Case Study: The "Empty Truck Syndrome" & Route Consolidation While clustering resolves the "spaghetti route" issue by enforcing strict geographic boundaries, it introduces a new inefficiency. Because clusters are solved as entirely isolated sub-problems, a vehicle might finish serving its assigned cluster with 40% capacity remaining, but is forced to return to the central depot instead of crossing into a neighboring cluster to continue deliveries. This leads to an excess of partially empty trucks traversing long distances to and from the depot.

To solve this, we implemented the `JoinRoutes` consolidation algorithm, which merges underutilized vehicles across different clusters. Figure 8 visually demonstrates the massive impact of this consolidation.



(a) Raw Clustering Output (b) Enhanced (JoinRoutes applied)

Figure 8: Visual comparison of route efficiency before and after JoinRoutes consolidation.

The visual and quantitative comparison (Table 1) reveals why a "Cluster-first, Route-second" approach requires a final consolidation step to reach global efficiency.

Metric	**Raw Clustering**	**Enhanced (JoinRoutes)**	**Improvement**
Total Fitness	34,637.35	**29,858.57**	~13.8% Reduction
Fleet Size	17 Vehicles	**14 Vehicles**	-3 Trucks Saved

Table 1: Quantitative performance impact of the JoinRoutes algorithm.

By merging routes, `JoinRoutes` eliminated redundant "to-and-from" depot trips, saving over 4,700 distance units. It successfully identified that single trucks could handle multiple boundary routes, creating much more continuous paths.

Phase 2 Conclusion: Geographic clustering is a fantastic tool for reducing spatial complexity and making the problem solvable for PSO. However, clustering alone is mathematically "selfish". The `JoinRoutes` consolidation is the essential bridge that transforms a collection of locally optimized neighborhoods into a globally efficient logistics plan.

5.1.4 Phase 3: Evaluation of Clustering Strategies

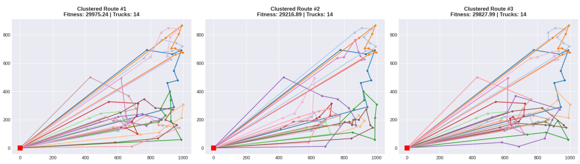
Having established the necessity of spatial partitioning and the `JoinRoutes` mechanism, this phase evaluates three distinct clustering strategies to optimize the sub-problem generation: Normal, Angular, and Complete.

1. Normal Clustering (Geographic & Demand): This approach employs the **K-Means** algorithm to partition the map using a normalized 3D feature vector: $(X_{norm}, Y_{norm}, \text{Demand}_{fake})$. Normalizing spatial coordinates to $[0, 1]$ ensures the raw scale of the map does not overshadow demand data. The Demand$_{fake}$ feature represents the absolute difference between a customer's load and the ideal vehicle average. This forces the clustering to group customers not just by location, but by how their loads complement each other to efficiently saturate a vehicle's capacity.

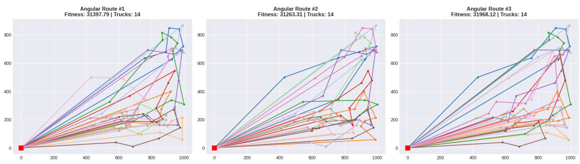
2. Angular Clustering (Polar Sweep): Shifting from Cartesian coordinates, this approach utilizes **polar coordinates** (azimuth) to group customers into wedge-shaped sectors radiating from the depot. This "Sweep Algorithm" topology naturally eliminates the spaghetti effect, as vehicles travel along specific directional vectors. However, being geometrically rigid, it is blind to customer demand constraints.

3. Complete Clustering (Hybrid): The ultimate synthesis of our spatial partitioning strategy. It blends angular boundaries with capacity/distance metrics, ensuring routes remain radially logical while simultaneously optimizing the internal load balancing of the vehicles.

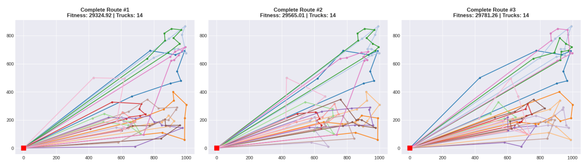
Figure 9 visualizes the Top 3 alternative routes generated by each strategy.



(a) Normal Clustering



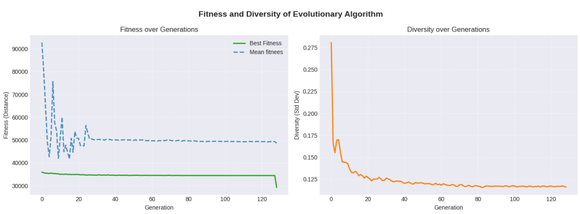
(b) Angular Clustering



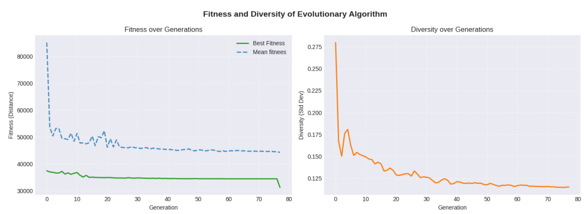
(c) Complete Clustering (Hybrid)

Figure 9: Comparison of the Top 3 alternatives extracted by each clustering strategy (post-JoinRoutes).

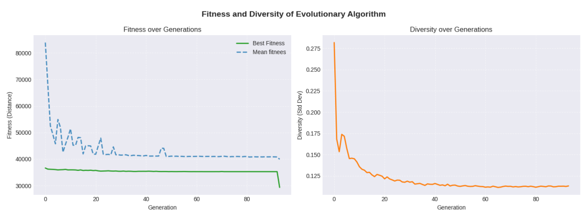
Convergence and Fitness Analysis: Unlike Phase 1, where fitness stagnated around 36,000, the clustered approaches drastically improved routing efficiency while maintaining exactly **14 trucks**. Figure 10 demonstrates highly aggressive convergence profiles tailored to sub-problem solving.



(a) Normal Clustering



(b) Angular Clustering



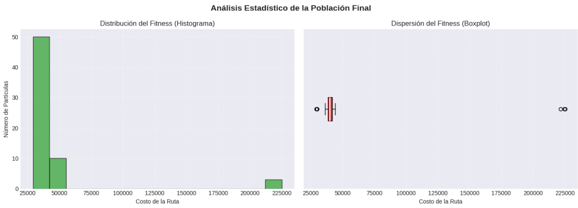
(c) Complete Clustering (Hybrid)

Figure 10: Evolution of Fitness and Diversity across the clustering strategies.

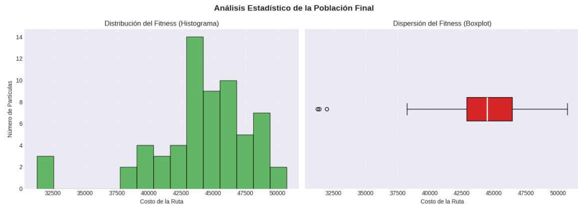
The **Angular** approach plateaued around 30,918, struggling slightly with its geometric in-flexibility. The **Normal** approach improved this to 29,259. However, the **Complete** approach emerged as the superior mathematical balance, dropping the global fitness to an impressive **29,195.53**. While this narrowly misses the theoretical 5% gap benchmark (27680) due to the inherent constraints of continuous-to-discrete SPV mapping, it represents a monumental optimization leap from the unclustered baseline.

Diversity and Population Statistics: In all three strategies, swarm diversity drops rapidly from an exploratory 0.28 to a highly stable state ($\sim$0.11). This absence of chaotic spikes proves the swarm efficiently focuses its search within the assigned logical boundaries, avoiding wasted evaluations on geographically distant nodes.

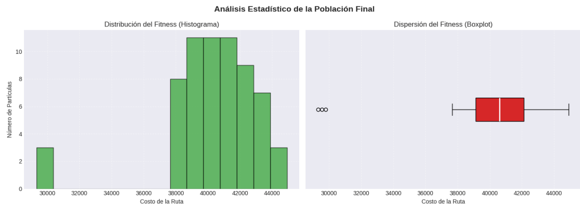
Figure 11 illustrates the statistical distribution of the final populations.



(a) Normal Clustering



(b) Angular Clustering



(c) Complete Clustering (Hybrid)

Figure 11: Histogram and Boxplot of the final population for each strategy.

The histograms reveal a primary concentration of successful particles in the 30,000–50,000 range. Notably, both Normal and Complete clustering exhibit distinct outliers near 225,000. These represent particles hit by the severe `truck_penalty` for failing to respect vehicle capacity, proving the algorithm aggressively discards weak logistical setups. The Complete approach maintains the tightest Interquartile Range, striking a superior balance compared to pure angular or K-Means methods.

To objectively quantify the final multimodal success, the Jaccard distances for the Top 3 routes were calculated:

- **Normal Clustering:** Alt 1 vs 2 (69.23%), Alt 1 vs 3 (67.78%), Alt 2 vs 3 (67.04%).
- **Angular Clustering:** Alt 1 vs 2 (68.51%), Alt 1 vs 3 (69.23%), Alt 2 vs 3 (72.04%).
- **Complete Clustering:** Alt 1 vs 2 (**73.40%**), Alt 1 vs 3 (**77.32%**), Alt 2 vs 3 (**76.68%**).

Phase 3 Conclusion: The **Complete Clustering** strategy is definitively the most robust architecture. By blending spatial sectorization with demand-packing, it successfully crushes the fitness baseline down to $\sim$29k, strictly limits the fleet to 14 trucks, and generates the highest structural diversity (Jaccard > 73%). It proves that the "Cluster-first, Route-second" paradigm, powered by a hybrid geometry-demand filter, is essential for tackling the Multimodal CVRP with Swarm Intelligence.

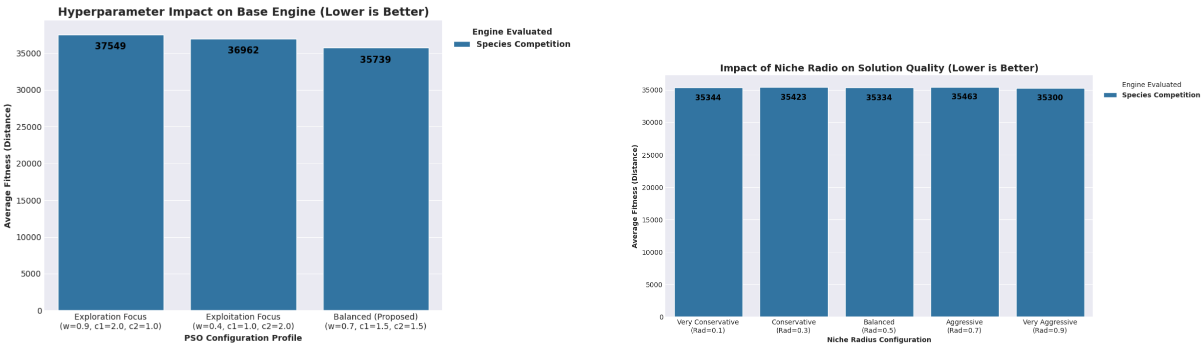
5.1.5 Phase 4: Hyperparameter Tuning

The performance of Particle Swarm Optimization is highly sensitive to its control parameters. A poorly tuned swarm will either prematurely converge into suboptimal local minima or scatter aimlessly. Rather than using arbitrary values, we implemented a dynamic tuning approach.

Static & Dynamic Baseline Components: The core swarm dynamics are balanced using an Inertia weight ($w = 0.7$) to maintain momentum, alongside equal Cognitive ($c_1 = 1.5$) and Social ($c_2 = 1.5$) coefficients. This ensures particles respect historical bests while being pulled toward the global optimum. Furthermore, computational effort is scaled dynamically based on topological complexity: instances with ≤ 200 nodes are limited to 5,000 evaluations to prevent overfitting, while larger maps are doubled to 10,000 evaluations.

With this baseline established, we conducted specific fine-tuning on the two superior architectures identified in the previous phases: *Species Competition* (Full-map) and *Complete Clustering* (Partitioned map).

1. Tuning the Species Competition Engine As concluded in Phase 1, Species Competition was the premier full-map algorithm due to its extraordinary Jaccard Structural Diversity ($> 91\%$) and parallel niche stability. Tuning focused on optimizing the w, c_1, c_2 coefficients and the Niche Radius (rad).



(a) Impact of PSO Coefficients

(b) Impact of Niche Radius

Figure 12: Hyperparameter tuning results for the Species Competition engine.

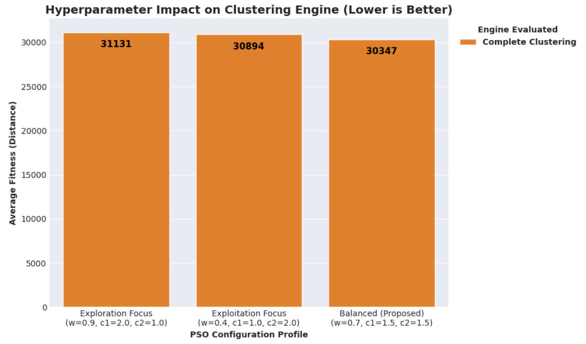
As observed in Figure 12, the **Balanced Profile** ($w = 0.7, c_1 = 1.5, c_2 = 1.5$) achieved the best fitness at **35,739**, proving significantly more effective than exploration-heavy or exploitation-heavy setups. Regarding the Niche Radius, narrower radii performed better. The Balanced (0.5) configuration reached the best result at **35,334**, while overly aggressive penalization degraded performance, likely collapsing swarm diversity.

2. Tuning the Complete Clustering Engine As concluded in Phase 3, Complete Clustering (combining X, Y with angular and complementary demand features) was the absolute best spatial technique, successfully reducing fitness below 30,000. Tuning focused on PSO coefficients alongside specialized clustering parameters: Niche Radius (rad), cluster density/size (K), and the truck excess penalty multiplier (P).

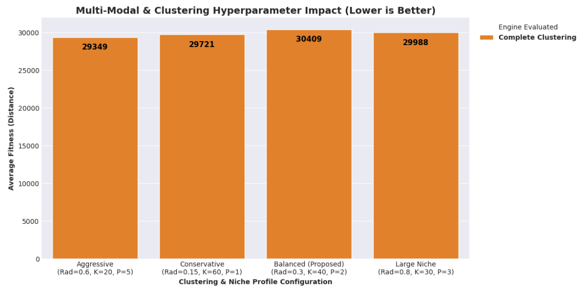
As seen in Figure 13, the **Balanced profile** again dominated the PSO movement dynamics, reaching a fitness of **30,347**. However, for the multimodal clustering parameters, the **Aggressive profile** ($rad = 0.6, K = 20, P = 5$) delivered the absolute best global fitness at **29,349**.

This reveals crucial insights into clustering mechanics:

- **Cluster Density (K):** Smaller, denser clusters ($K = 20$) allowed the localized PSO to find high-precision sub-routes more effectively than larger geographic boundaries ($K = 60$).
- **Truck Penalty (P):** A severe truck penalty ($P = 5$) forced the algorithm to relentlessly optimize vehicle saturation, resulting in superior fleet management.
- **Dynamic Radius Consistency:** By scaling the Niche Radius with the problem's physical dimension ($\sqrt{N}$), the "Aggressive" clustering profile succeeded where the base engine's aggressive settings previously failed, preventing the "fitness flattening" effect.



(a) Impact of PSO Coefficients



(b) Impact of Clustering Parameters (Rad, K, P)

Figure 13: Hyperparameter tuning results for the Complete Clustering engine.

Phase 4 Conclusion: The tuning process confirms that while standard PSO movement coefficients prefer a balanced 0.7/1.5/1.5 profile across all architectures, the multimodal parameters require distinct treatments. Full-map strategies require conservative niche radii to avoid search space collapse, whereas partitioned clustering architectures thrive under aggressive geometric and penalization constraints to force maximum vehicle saturation within small neighborhoods.

5.1.6 Phase 5: Statistical Significance Testing

While individual executions and visual maps strongly suggest that the **Complete Hybrid Clustering PSO** outperforms the Baseline and Species Competition methods, empirical observations are insufficient to claim algorithmic superiority due to the stochastic nature of evolutionary algorithms. To definitively prove the robustness of our proposed architecture, a rigorous statistical benchmark was conducted.

Methodology: We executed 35 independent runs for each algorithmic approach across 21 random maps of the dataset (but finally, we kept with 52). The runs were aggregated by extracting the median performance metric for each specific map to eliminate intrinsic instance noise. Because fitness distributions of evolutionary algorithms rarely follow a perfect Gaussian distribution, we employed the non-parametric **Friedman Aligned Ranks test** to evaluate global differences ($\alpha = 0.05$), followed by the **Shaffer post-hoc test** for adjusted pairwise comparisons.

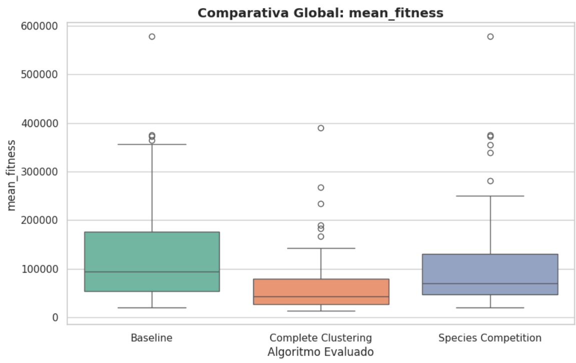
Figure 14 visualizes the distribution of the six key metrics evaluated across the dataset.

Based on the rigorous Shaffer post-hoc procedure, we draw the following definitive conclusions:

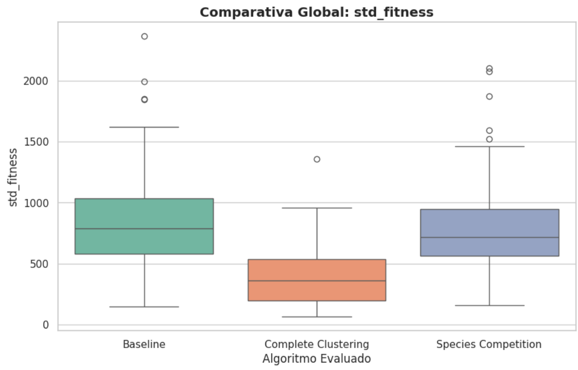
- **Superiority in Quality and Stability:** The Friedman test indicated significant global differences. The Shaffer post-hoc confirmed that *Complete Clustering* is statistically superior ($p_{adj} < 0.001$) to both Baseline and Species Competition in minimizing routing cost and standard deviation. It is vastly more accurate and robust.

- **The Search Effort Paradox:** A compelling insight emerges regarding efficiency. While Complete Clustering requires a statistically higher number of evaluations and generations to converge, its actual execution time in seconds is statistically **lower (faster)** than the alternatives ($p_{adj} < 0.001$). Decomposing the map into clusters drastically reduces the dimensional complexity of the internal PSO matrix operations, allowing a deeper search in a fraction of the clock time.

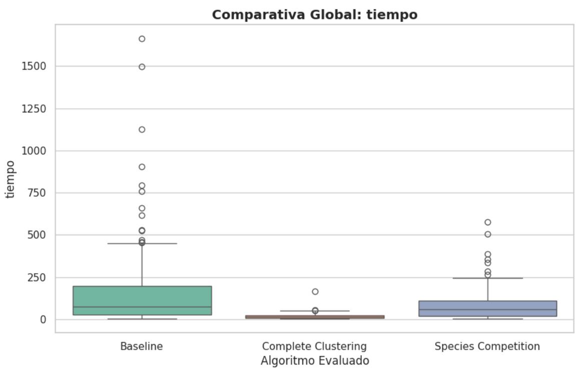
- **Unproductive Diversity vs. Exploitation:** Baseline and Species Competition maintained significantly higher structural diversity (Jaccard). However, given their inferior



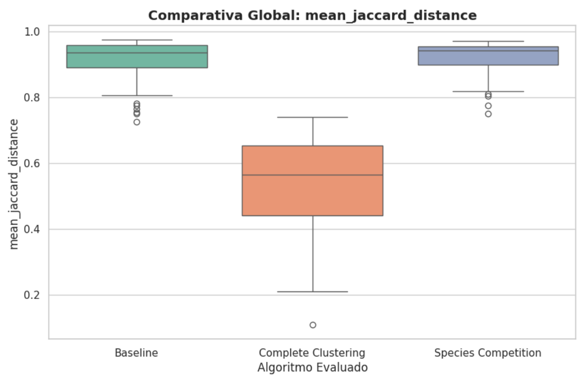
(a) Routing Cost (mean_fitness)



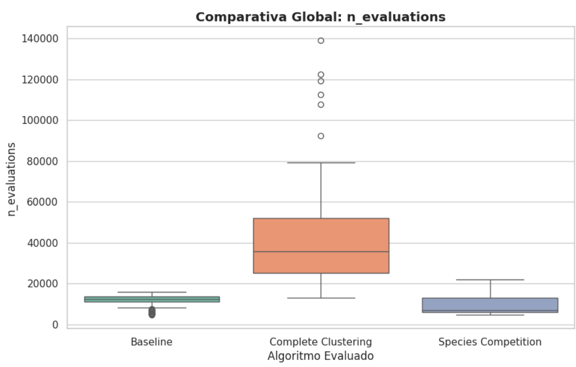
(b) Stability (std_fitness)



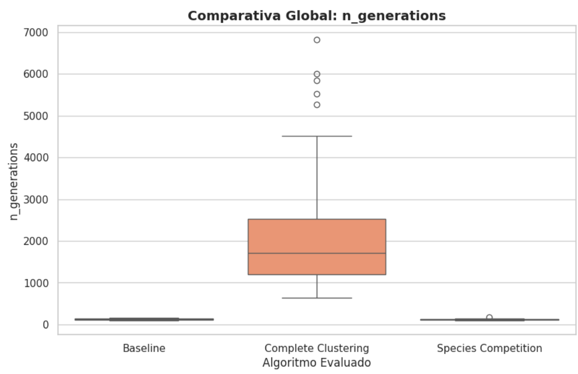
(c) Execution Speed (tiempo)



(d) Diversity (mean_jaccard)



(e) Effort (n_evaluations)



(f) Effort (n_generations)

Figure 14: Global statistical comparison of the three primary PSO architectures.

fitness, this indicates they are wandering inefficiently through unfeasible search spaces. Complete Clustering sacrifices global diversity to heavily exploit valid routing zones.

- **Ineffectiveness of Species Competition:** Across all six evaluated metrics, Species Competition failed to demonstrate any statistically significant difference when compared to the standard Baseline PSO (statistical tie). The isolated addition of species competition does not provide a tangible benefit for this discrete combinatorial problem.

Statistical Verdict: The evidence unequivocally rejects the null hypothesis. The integration of the clustering architecture transforms the PSO into a significantly faster and highly optimized solver for the CVRP, while pure continuous approaches fail to compete efficiently.

5.2 PSO Overall Conclusions

The comprehensive evaluation of the CVRP resolution through PSO variants provides a definitive roadmap of what succeeds in high-dimensional routing optimization:

1. **Structural Innovation vs. Fine-Tuning:** Early stages proved that while fine-tuning (w, c_1, c_2) is necessary for stability, it cannot overcome the discrete nature of the CVRP. The most significant performance gains came from re-engineering the architecture through **Clustering**.

2. **The Impact of Heuristic Operators:** Local search (2-opt) proved indispensable for refining individual routes, compensating for the swarm's lack of geometric "fine motor skills". The `JoinRoutes` algorithm was equally essential for constraint satisfaction and fleet reduction.

3. **Limitations in Highly Constrained Scenarios:** While Clustering PSO dominates in routing cost, statistical evidence shows it occasionally struggles with absolute truck count constraints in highly complex maps. The `JoinRoutes` logic can reach a plateau where merging further routes violates capacity limits, highlighting a bottleneck in pure structural partitioning.

Table 2 summarizes the architectural capabilities.

Feature	**Baseline PSO**	**Species Competition**	**Complete Clustering**
Solution Quality	Moderate	Moderate	**Excellent**
Execution Speed	Slow	Slow	**Very Fast**
Stability (Std)	Low	Low	**High**
Diversity	High	High	Moderate
Implementation	Simple	Complex	Highly Complex

Table 2: Final comparative summary of the evaluated PSO architectures.

Final Verdict: The integration of structural Complete Clustering and local heuristics transforms the PSO from a generic continuous optimizer into a specialized CVRP solver. The results unequivocally demonstrate that structural hybridization is the only viable path for resolving complex multimodal logistical constraints efficiently.

5.3 ACO Internal Comparison

Ant Colony Optimization is naturally designed for combinatorial routing problems like the CVRP. Artificial ants probabilistically construct solutions by traversing a graph, navigating from node to node. Their decisions are guided by a combination of **artificial pheromone**

trails (the collective learned experience of the swarm) and **heuristic visibility** (the inverse of the physical distance). This native graph-building approach makes ACO highly intuitive and extremely effective for spatial pathfinding.

To strictly validate and benchmark our ACO architectures, we utilized the same **X-n106-k14** benchmark instance as the PSO (with a known optimal fitness of **26362** and a 5% multimodal quality threshold of **27680**).

5.3.1 Experimental Design and Constraint Dynamics

We implemented a highly modular ACO engine evaluating a 2×2 experimental design based on two key dimensions:

1. **Constraint Handling:** *Feasibility* (hard constraints physically preventing capacity overloads) versus *Penalty* (soft constraints allowing invalid routes but applying massive mathematical penalties to the final fitness).

2. **Multimodal Techniques:** *Archive Niching* (passive background saving of diverse routes) versus *Sequential Niching* (active erasure of pheromone trails to force temporary amnesia).

Computational Note: Unlike the PSO methodology, computing the Jaccard structural diversity across the entire ant colony at every single iteration introduces an unacceptable computational bottleneck. Therefore, continuous diversity evolution curves over time are omitted for the ACO. Structural diversity is strictly evaluated on the final archived niches.

5.3.2 Phase 1: The Archive Method with Feasibility Constraints

The most traditional way to handle a combinatorial problem is to strictly enforce its constraints. In this approach, we apply a **Feasibility** rule: an ant is physically blocked from adding a customer to its route if the customer's demand exceeds the remaining capacity of the vehicle. This acts as a hard "wall," ensuring that the swarm only evaluates 100% valid routes.

To achieve multimodality, we use **Passive Archiving**. The ACO runs its normal course, but a background mechanism collects all solutions that fall within a defined quality threshold. At the end of the execution, this archive is filtered using the Jaccard distance to return only solutions that are structurally distinct.

Figure 15 visualizes the Top 3 structurally diverse alternatives extracted by this method.

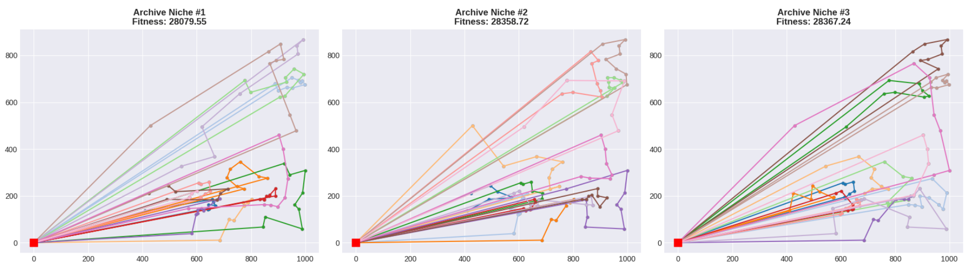
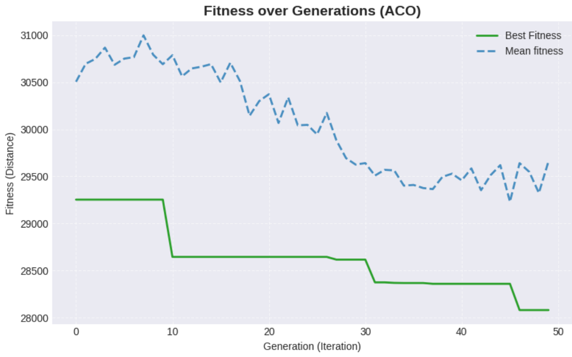
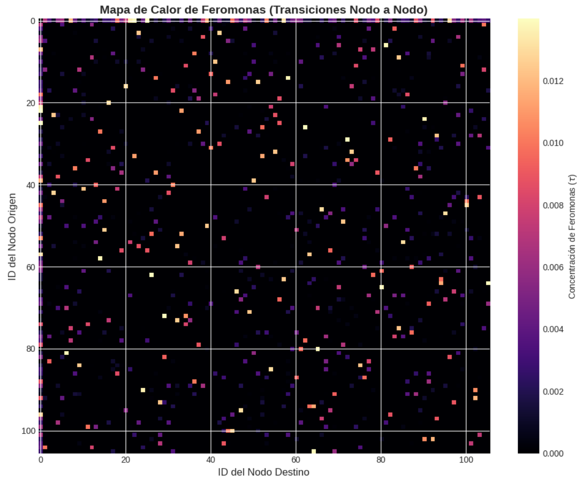


Figure 15: Top 3 alternative routes discovered using Archive Niching with Feasibility constraints.

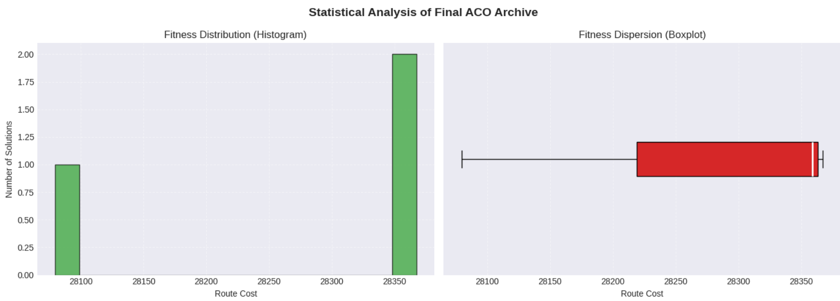
Figure 16 illustrates the internal swarm learning dynamics and the final distribution of the archived population.



(a) Fitness Convergence



(b) Pheromone Consolidation Matrix



(c) Distribution of the Final Archived Population

Figure 16: Performance metrics and swarm dynamics for Feasibility + Archive.

Analysis and Conclusions:

- **Convergence and Fitness:** The fitness plot displays a clear, stepped minimization for the Best Fitness, successfully stabilizing near the 28,000 mark. Interestingly, the Mean Fitness remains highly variable. This gap indicates a healthy swarm dynamic: while elite ants exploit the best paths, the passive nature of this configuration allows a large portion of the colony to continuously explore, preventing total premature convergence.
- **Pheromone Consolidation:** The Pheromone Matrix Heatmap visually confirms the swarm's learning behavior. The vast majority of the matrix is dark (evaporated paths), with a few bright pixels representing the specific optimal "highways" the colony definitively agreed upon.
- **Archive Filtering and Diversity:** The final Jaccard distances (**69.95%**, **72.73%**, and **72.73%**) vastly exceeded our 20% minimum diversity requirement. Since we are using a Pure ACO approach without a heavy global 2-opt post-processing, some paths naturally cross, but the structural uniqueness of each niche guarantees highly diverse real-world contingency options.

5.3.3 Phase 2: The Archive Method with Adaptive Penalty Constraints

Hard constraints (Feasibility) can sometimes trap the swarm in local optima by creating invisible walls on the map, forcing vehicles to return to the depot prematurely. To test this hypothesis, we introduced **Soft Constraints via Adaptive Penalties**.

Ants are now allowed to visit any customer and overload their trucks, building physically invalid routes during construction. However, during evaluation, a massive penalty factor ($2 \times \sum$ Distance) is applied based on the excess weight. This allows the swarm to temporarily explore "forbidden zones," relying on pheromone evaporation to naturally phase out overloaded routes.

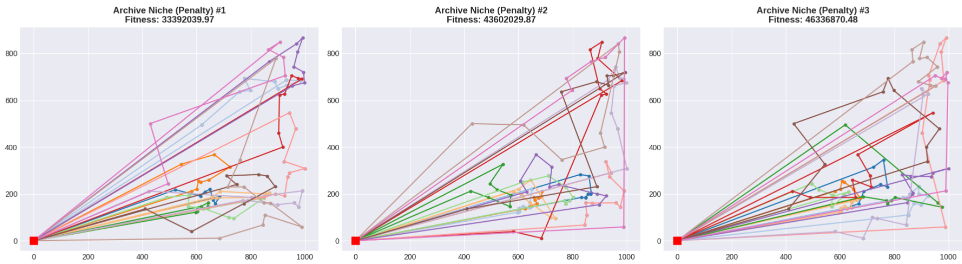
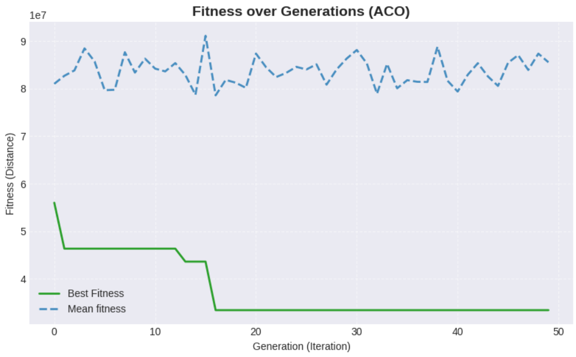
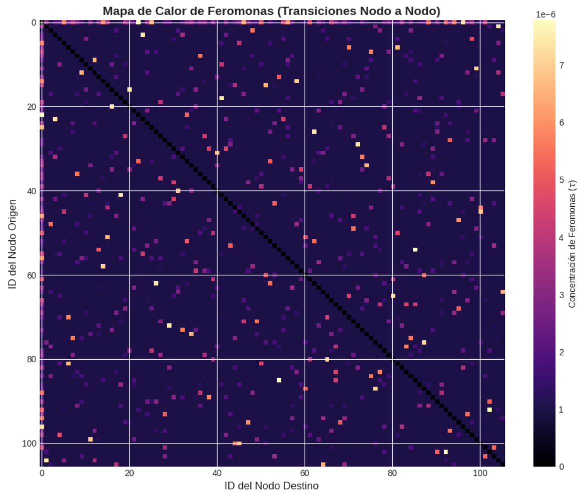


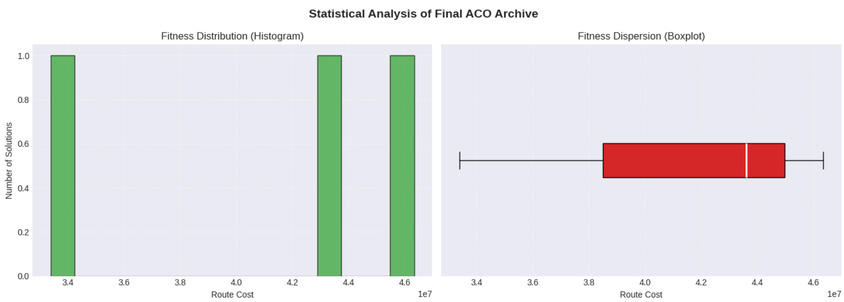
Figure 17: Top 3 alternative routes discovered using Archive Niching with Penalty constraints.



(a) Fitness Explosion



(b) Diluted Pheromone Matrix



(c) Distribution of the Heavily Penalized Population

Figure 18: Performance metrics and swarm dynamics for Penalty + Archive.

Analysis and Conclusions:

- **Fitness Explosion:** The convergence plot reveals a drastic shift. The Best Fitness stabilizes at an extraordinarily high value ($\sim 33,000,000$), while Mean Fitness fluctuates near $85,000,000$. The swarm heavily built invalid routes, triggering massive penalties, and failed to naturally phase them out within the allotted generations.
- **Pheromone Dilution:** The Heatmap shows maximum chemical concentration (τ) on the

scale of 10^{-6}, significantly lower than the Feasibility approach. Because explored routes were heavily penalized, the deposited pheromone (Q/cost) was infinitesimal. The swarm failed to establish strong "highways", leading to a highly diluted learning process.

- **Extreme Diversity vs. Quality:** Jaccard distances were remarkably high (**79.59% to 84.31%**). By allowing ants to ignore capacity walls, they freely explored completely different topologies. However, this extreme structural diversity came at the absolute cost of solution quality.

Phase 1 & 2 Conclusion: This experiment empirically demonstrates that while Adaptive Penalties succeed in generating diversity, they are highly inefficient for a pure ACO in the CVRP context. The lack of hard boundaries prevents the rapid consolidation of valid routes. Therefore, Feasibility rules are strictly required to guide heuristic visibility effectively.

5.3.4 Phase 3: The Sequential Method with Feasibility Constraints

While Passive Archiving relies on the natural exploration of the swarm to find different routes before they converge, **Sequential Niching** forces it actively. Execution is divided into intervals. When an interval finishes, the best route is stored as a valid niche, and immediately after, a massive "pheromone penalty" is applied: the optimal trail is actively evaporated (multiplied by 0.0001).

When the next interval begins, the ants' favorite path has essentially vanished from their collective memory, forcing them to explore entirely new geographical zones. This mathematically guarantees structural diversity. We combine this active memory erasure with the **Feasibility** rule (hard constraints) to ensure all newly explored areas remain valid and capacity-compliant.

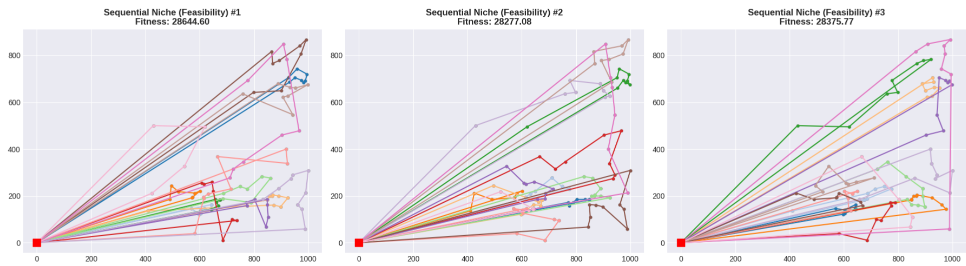
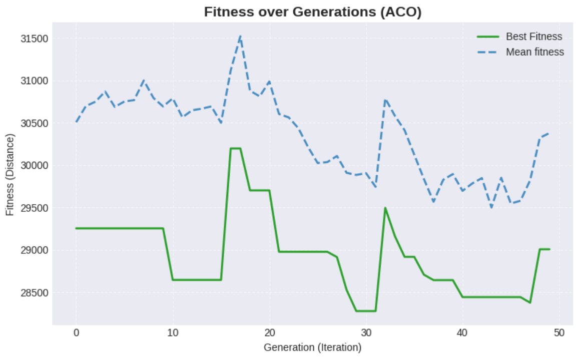
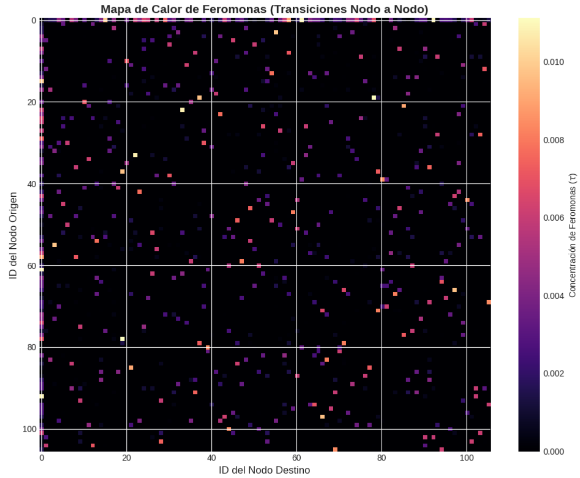


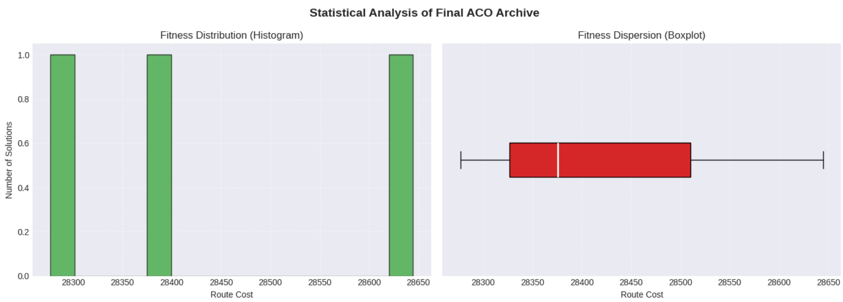
Figure 19: Top 3 alternative routes discovered using Sequential Niching with Feasibility constraints.



(a) Forced Amnesia Convergence



(b) Sequential Pheromone Matrix



(c) Distribution of the Sequentially Archived Population

Figure 20: Performance metrics and swarm dynamics for Feasibility + Sequential.

Analysis and Conclusions:

- **Forced Amnesia and Convergence:** The fitness plot perfectly illustrates the active memory erasure mechanic. Distinct, abrupt spikes in fitness represent the exact moments the pheromone trails were heavily evaporated. The swarm experiences forced "amnesia," temporarily degrading its performance before rapidly converging into a new, distinct local optimum.
- **Pheromone Consolidation:** Unlike the penalty approach, the Heatmap confirms a healthy chemical deposition (scale of $\sim$ 0.010). Multiple bright spots across different regions reflect the sequential discovery of distinct "highways" during each interval.
- **Active Structural Isolation:** Jaccard distance metrics represent a significant upgrade over the passive Archive method. By actively punishing known routes, the algorithm secured structural diversities ranging from **75.39% to 81.59%**. The swarm engineered three entirely distinct logistical strategies within the highly competitive 28,200–28,650 fitness range, showcasing the supreme effectiveness of combining Feasibility with Sequential Niching.

5.3.5 Phase 4: The Sequential Method with Adaptive Penalty Constraints

Finally, we test the most aggressive and chaotic configuration by combining the two active disruption techniques: **Soft Constraints (Adaptive Penalties)** and **Sequential Niching (Active Memory Erasure)**.

Ants are granted absolute spatial freedom—they can traverse any edge and ignore capacities. However, their overloads are heavily penalized mathematically. Furthermore, any optimum they manage to consolidate under these harsh conditions is actively erased at specific intervals. This tests whether total exploratory freedom combined with forced amnesia yields better structural diversity than strictly restricting their physical movements.

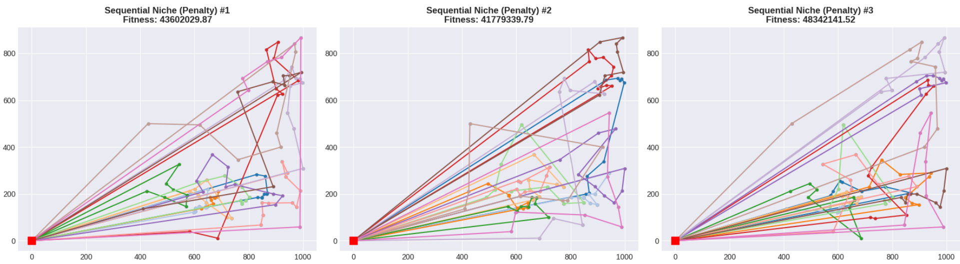
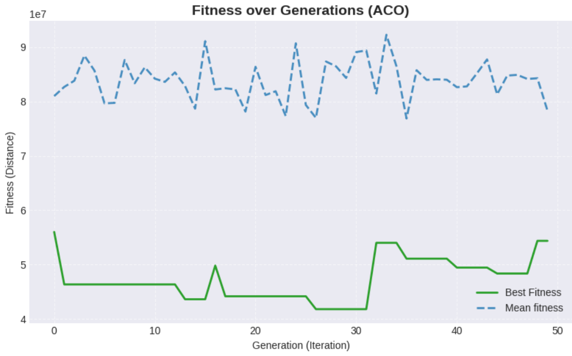
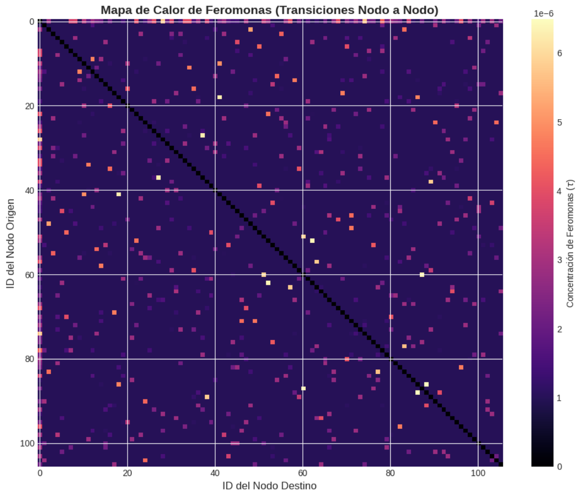


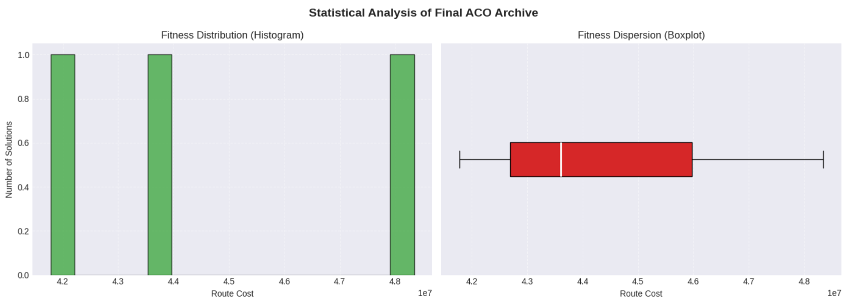
Figure 21: Top 3 alternative routes discovered using Sequential Niching with Penalty constraints.



(a) Chaotic Swarm Disruption



(b) Pheromone Starvation Matrix



(c) Distribution of the Invalid Penalized Population

Figure 22: Performance metrics and swarm dynamics for Penalty + Sequential.

Analysis and Conclusions:

- **Total Swarm Disruption:** The fitness plot displays absolute chaos. Best Fitness fluctuates wildly between 40M and 50M. The sequential memory erasures completely destabilize a swarm that is already struggling to find valid routes due to the lack of physical constraints.
- **Pheromone Starvation:** Because every route is heavily overloaded and penalized into the millions, the pheromone deposited is practically zero (10^{-6}). Combined with active

erasure, the swarm suffers from "pheromone starvation", never building a strong memory to guide subsequent generations.

- **Maximum Diversity at the Cost of Viability:** Jaccard metrics reached a staggering **93.69%**. However, the Alternative Route Maps reveal chaotic, highly crossed "spaghetti" routes composed purely of highly penalized, overloaded contingencies.

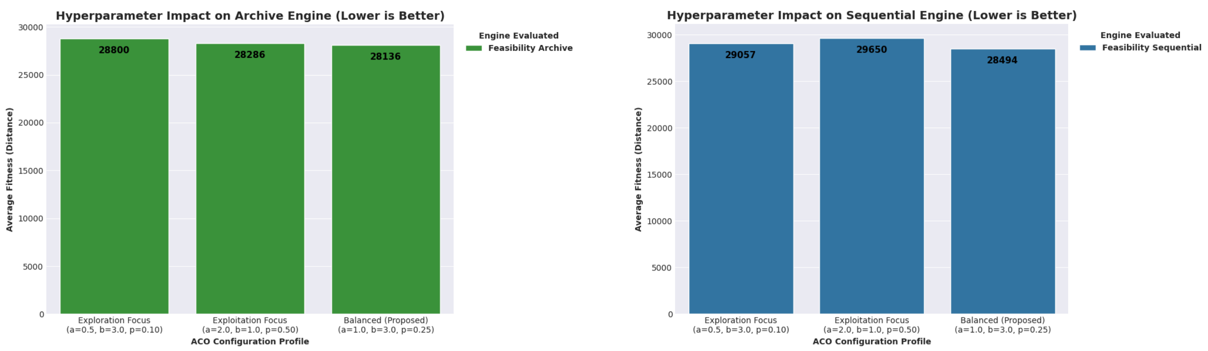
ACO Architecture Verdict: This final experiment serves as a crucial proof of concept. Stripping away hard boundaries while simultaneously destroying the swarm's collective memory breaks the core heuristic engine of the ACO. We definitively conclude that for the CVRP, penalty-based constraints are unviable. **Feasibility combined with Sequential Niching or Passive Archiving** are the only superior multimodal strategies, and will advance to the hyperparameter tuning phase.

5.3.6 Phase 5: Hyperparameter Tuning

The performance of Ant Colony Optimization is highly sensitive to the balance between exploration and exploitation. Rather than using arbitrary values, a dynamic tuning approach was implemented. Computational effort (ants and iterations) is scaled dynamically based on topological complexity to ensure high-density maps (> 200 nodes) receive sufficient depth to resolve structural routing.

Tuning focused strictly on the two viable constraints identified: **Feasibility + Archive** and **Feasibility + Sequential**. The objective was to calibrate the Pheromone weight (α), Heuristic weight (β), and Evaporation rate (ρ), alongside the specific multimodal thresholds.

1. Tuning Core Heuristics (Archive vs. Sequential) Figure 23 illustrates the impact of adjusting the α, β, and ρ parameters on both the passive and active memory architectures.



(a) Hyperparameter Impact (Archive Engine)

(b) Hyperparameter Impact (Sequential Engine)

Figure 23: Comparison of ACO transition rule profiles across both Feasibility engines.

In contrast to initial expectations, both multimodal engines performed absolute best under the **Balanced Profile** ($\alpha = 1.0, \beta = 3.0, \rho = 0.25$). This configuration provides the optimal equilibrium between greedy heuristic visibility and pheromone persistence.

Crucially, the data confirms that the passive **Archive method consistently outperformed** the active Sequential method across all tested profiles. Comparing their peaks, the Archive method's best fitness (**28,194**) is significantly better than the Sequential method's best (**28,581**). This definitively proves that "nuking" the pheromone matrix in the Sequential approach to force amnesia incurs a measurable cost in convergence quality.

2. Tuning the Archive Diversity Threshold Having established the *Feasibility + Archive* engine as the superior baseline, we fine-tuned its specific multimodal parameter: the `min_diversity` threshold (Jaccard distance) required to accept a route into the archive.

As seen in Figure 24, tuning the Jaccard requirement reveals a clear U-shaped curve in quality. The **Aggressive (Div=0.30)** profile achieved the absolute best global fitness at **28,129**.

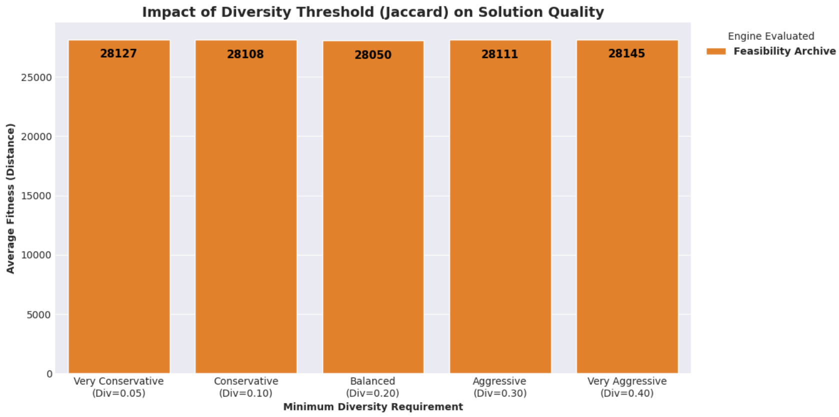


Figure 24: Impact of the Jaccard Diversity Threshold on solution quality (Archive Engine).

- **Redundancy Limit:** Lower thresholds (e.g., 0.10) scored worse (28,183), suggesting that allowing too much structural redundancy prevents the swarm from isolating the most geometrically efficient paths.
- **Isolation Limit:** Conversely, pushing the requirement to 0.40 degraded fitness to 28,155. When structural diversity is forced beyond a natural equilibrium, the algorithm is compelled to accept sub-optimal, longer routes simply to satisfy the diversity constraint.

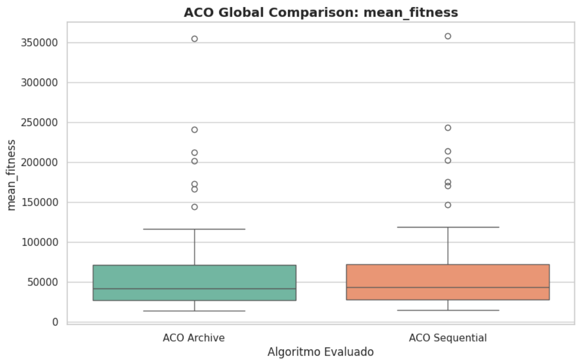
Phase 5 Conclusion: The empirical data dictates that CVRP optimization via ACO requires the Balanced heuristic setup ($\alpha = 1.0, \beta = 3.0, \rho = 0.25$). For absolute peak performance, the **Feasibility + Archive** engine paired with a **30% minimum Jaccard requirement** stands as the superior strategy, delivering the best fitness found in the ACO evaluation phase.

5.3.7 Phase 6: Statistical Significance Testing

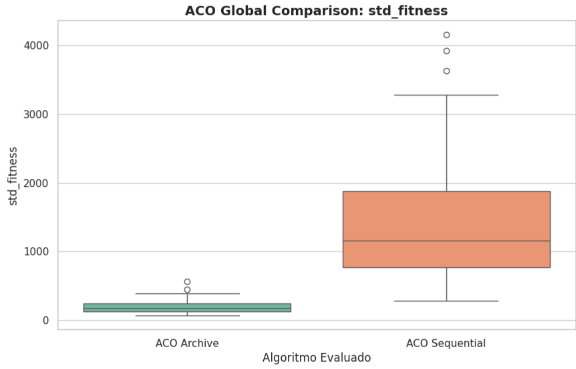
While individual executions strongly suggest that the **Feasibility Archive ACO** outperforms the Sequential method, empirical observations are insufficient to claim algorithmic superiority. Evolutionary algorithms are stochastic by nature; a good result could simply be a statistical anomaly.

To definitively prove the robustness of our architecture, we conducted a rigorous statistical benchmark. We executed 35 independent runs for each algorithmic approach across 52 random maps of the dataset. We aggregated the runs by extracting the median performance metric for each specific map to eliminate intrinsic noise. Because we are comparing exactly two independent algorithms simultaneously (the viable Feasibility variants), we employed the non-parametric **Wilcoxon signed-rank test** to evaluate differences at a strict 95% confidence level ($\alpha = 0.05$).

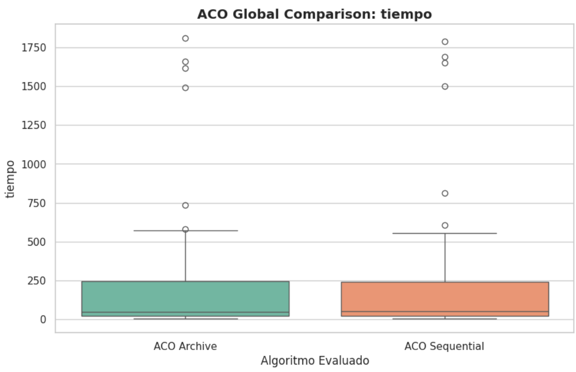
Figure 25 visualizes the global distribution of the four evaluated metrics. *Note: Computational effort metrics (evaluations and generations) are omitted as they are statically fixed across all ACO runs.*



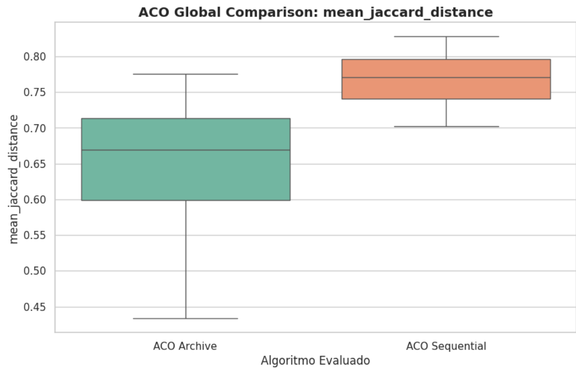
(a) Routing Cost (mean_fitness)



(b) Stability (std_fitness)



(c) Execution Speed (tiempo)



(d) Diversity (mean_jaccard)

Figure 25: Global statistical comparison between the Archive and Sequential ACO architectures.

Based on the Wilcoxon signed-rank test, we draw the following definitive conclusions:

- **Superiority in Solution Quality and Stability:** The *Feasibility Archive ACO* is statistically superior ($p < 0.001$) to the Sequential variant in terms of minimizing the routing cost. Furthermore, it demonstrates a significantly lower standard deviation, proving that the passive archive mechanism is not only more accurate but also far more robust and consistent across multiple stochastic executions.

- **Computational Efficiency:** The statistical tests reveal a technical tie in computational time ($p \approx 0.150$). The computational overhead of the Sequential method's active pheromone erasure proved to be negligible compared to the Archive's mathematical filtering, meaning both multimodal engines are equally fast.

- **The Cost of Active Diversity:** As hypothesized, the Sequential ACO maintains a significantly higher population diversity ($p < 0.001$). However, given its inferior fitness results, this proves that active destruction of the pheromone matrix (induced amnesia) forces the colony to explore sub-optimal regions. The Archive approach sacrifices extreme diversity to heavily exploit valid routing zones.

Statistical Verdict: The statistical evidence unequivocally proves that for CVRP optimization using ACO, preserving the continuous evolution of the pheromone matrix while passively archiving structurally distinct solutions is mathematically superior to active memory erasure.

5.4 ACO Overall Conclusions

The comprehensive evaluation of the CVRP resolution through ACO variants provides a definitive roadmap of what works and what fails in bio-inspired routing optimization:

1. **Constraint Handling is Paramount:** The most significant insight is the overwhelming superiority of **Feasibility-based search** over Penalty-based methods. In ACO, where learning depends on the inverse of the cost, high-penalty "soft walls" destroy the feedback loop (Pheromone Starvation), proving that strict physical boundaries are the only viable path for swarm-based routing.

2. **The Impact of Heuristic Bias and Memory:** The optimal **Balanced Profile** ($\beta = 3.0, \alpha = 1.0, \rho = 0.25$) consistently outperformed exploration-heavy setups. The spatial information of the map (heuristic visibility) is a highly reliable guide, while the 25% evaporation rate perfectly balances the fading of sub-optimal early trails with the consolidation of a strong collective memory.

3. **Diversity vs. Optimization:** The Sequential variant maintains higher population diversity, but this diversity is "unproductive" (inferior fitness). The Archive ACO intentionally focuses the swarm's energy on high-quality regions (Exploitation over Exploration) while relying on end-of-run mathematical filtering to extract valid contingency plans.

4. **Algorithmic Limitations:** While the Archive ACO is highly robust, pure swarm variants show increased difficulty in scenarios with extreme node density. As the discrete search space grows exponentially, ensuring absolute precision in the final node sequences pushes the limits of pure swarm intelligence without massive computational resources.

Table 3 summarizes the architectural capabilities.

Feature	**Sequential ACO**	**Archive ACO**
Solution Quality	Good	**Excellent**
Execution Speed	Fast (Tie)	Fast (Tie)
Stability (Std)	Moderate	**High**
Diversity (Jaccard)	**Highest**	Moderate
Swarm Learning	Active (Erasure)	**Passive (Preserved)**

Table 3: Final comparative summary of the evaluated ACO architectures.

Final Verdict: The integration of a feasibility-based passive archive transforms the ACO into a highly specialized and robust CVRP solver. The results unequivocally show that **respecting the natural evolution of pheromone trails** while applying **strict capacity boundaries** is the mathematically superior path for solving complex multimodal logistical constraints efficiently.

5.5 GA Internal Comparison

To systematically evaluate the evolutionary robustness of the Genetic Algorithm, we conducted a five-phase experimental progression. This analysis transitions from a baseline feasibility study to a highly exploited memetic architecture utilizing explicit niching.

Following the requirements, all phases were validated using the **X-n106-k14** benchmark. The experiments aim to overcome the "Heuristic Stagnation" observed in pure evolutionary searches by integrating local search refinement and niche-replacement logic.

Genetic Algorithm (GA) Implementation Mechanisms

To elevate the Genetic Algorithm from a basic combinatorial search to a highly competitive multimodal engine, we implemented several advanced operational mechanics prior to the experimental phase:

- **Heuristic Seeding (The 80/20 Rule):** Generating a purely random initial population (Generation 0) for the CVRP creates immense structural noise, resulting in highly inefficient, crisscrossing routes. To provide the GA with a competitive "warm start," chromosomes are initialized using a *Randomized Nearest Neighbor* heuristic. To prevent premature convergence, an 80/20 split is enforced: 80% of the initial population is heuristically guided, while 20% remains purely random. This preserves a vital reservoir of chaotic genetic material for the crossover operators to exploit during later generations.
- **Memetic refinement (Windowed 2-opt):** Pure evolutionary operators (such as OX1 Crossover) frequently leave geometric knots within individual vehicle sub-routes. To resolve this, the GA is hybridized into a **Memetic Algorithm** utilizing a *Windowed 2-opt Local Search*. By restricting the edge-untangling process to a localized sliding window rather than the entire global map, the algorithm avoids severe computational overhead ($\mathcal{O}(N^2)$), achieving a highly efficient balance between evolutionary exploration and localized spatial exploitation.
- **Multimodal Architectures:** To satisfy the requirement of extracting three distinct routing topologies, the GA's replacement and memory mechanics were heavily modified. We implemented both passive niching (**Deterministic Crowding / Archive Method**), which utilizes a Jaccard distance threshold to maintain diverse "species" in parallel, and active niching (**Sequential Tabu**), which explicitly bans previously discovered high-efficiency edges to force exploration into new geographical sectors.

5.5.1 Phase 1: Pure GA vs. Memetic Analysis (The Engine)

This initial phase established the baseline performance by contrasting a standard evolutionary search against the memetic hybrid, quantifying the empirical impact of localized refinement.

Case Study: "Heuristic Stagnation" & Memetic Healing While our Cluster-First heuristic seeding resolves the initial "blind start" issue by creating geometrically logical routes, it introduces a new challenge for pure evolutionary operators. Because the initial seeds are already highly optimized, standard crossover operators (like OX1) inherently disrupt their local structure. Without a geometric repair mechanism, the generated offspring are strictly worse than their parents and are instantly discarded by tournament selection. This causes the Pure GA to stagnate immediately at the Generation 0 baseline.

To overcome this, we introduced a **Windowed 2-opt** memetic operator. Instead of allowing the swarm to blindly rely on genetic recombination, this localized search acts as a healing step, systematically untangling crossed edges in the newly generated offspring before their fitness is evaluated. Figure 26 visually demonstrates the impact of this memetic refinement.

The visual and quantitative comparison (Table 4) reveals why a pure evolutionary approach requires a local search repair step to reach global efficiency in highly constrained CVRP spaces.

Metric	Pure GA (Baseline)	Enhanced (Memetic GA)	Improvement
Total Fitness	27,493.15	**27,424.11**	∼69.04 Reduction
Execution Time	54.10s	64.37s	+10.27s Overhead

Table 4: Quantitative performance impact of the Memetic 2-opt local search.

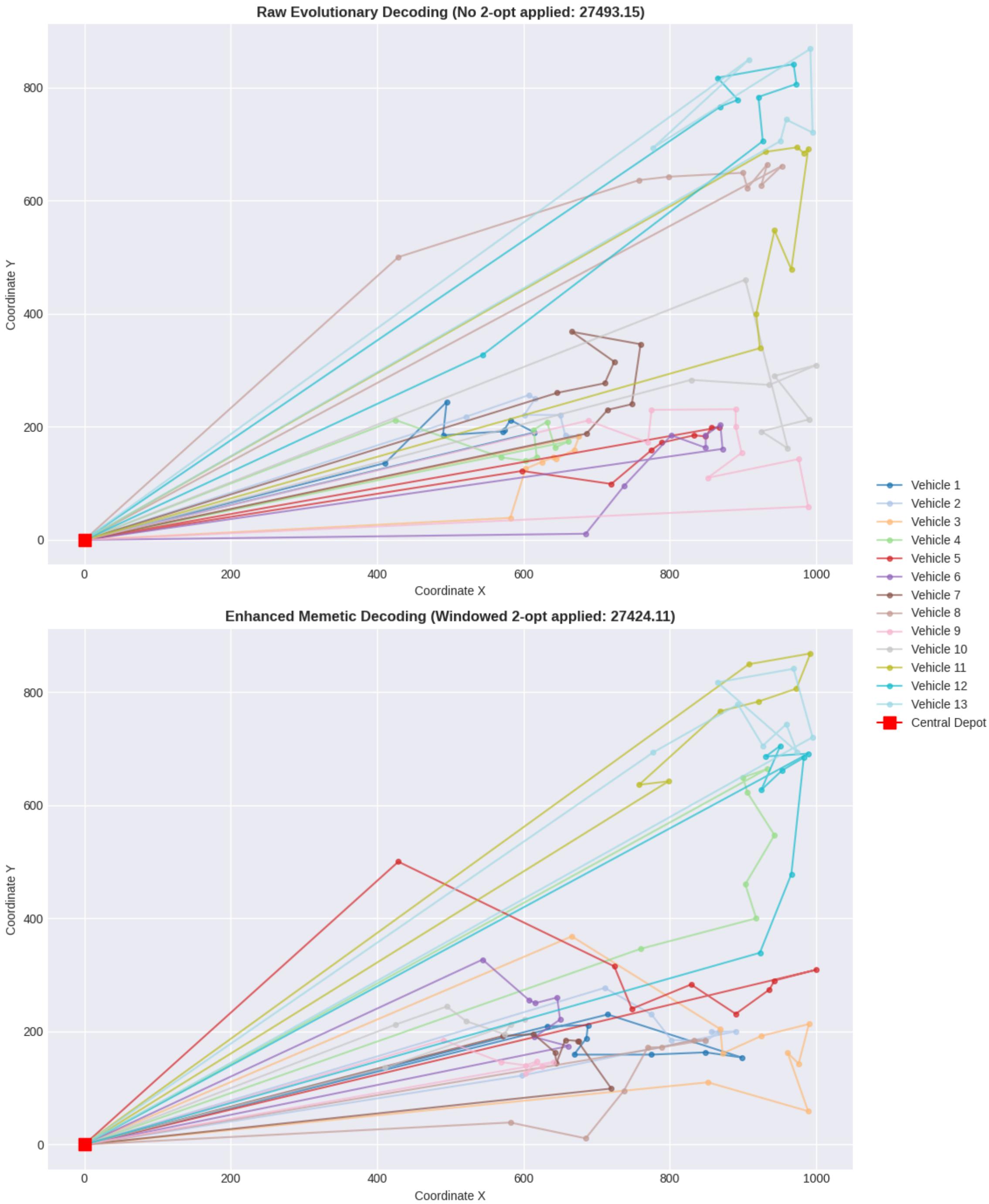


Figure 26: Visual comparison of route efficiency: Raw Evolutionary Search (Pure GA) vs. Enhanced Search (Memetic 2-opt applied).

5.5.2 Phase 2: Deterministic Crowding and Multimodal Extraction

To extract three diverse routing alternatives, we had to overcome *Genetic Drift*—the tendency of standard GA selection pressure to force premature convergence onto a single, unimodal path.

Case Study: Active Niching via Deterministic Crowding

We altered the survival mechanics by implementing **Deterministic Crowding**. When a new offspring is generated, its topology is compared against the population using **Jaccard Distance**. If it falls below a `similarity_threshold` of 0.20, it competes specifically against its closest structural relative. This creates a hard mathematical boundary against genetic homogenization.

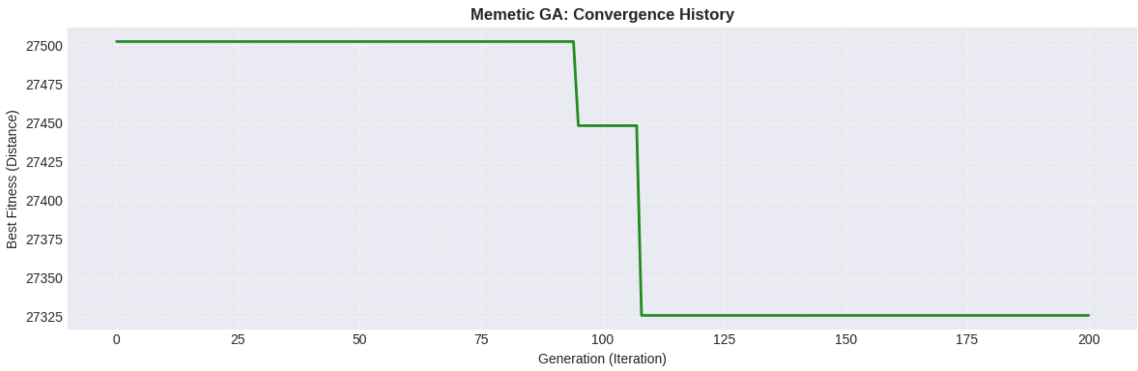


Figure 27: Phase 2 Results: Convergence history (top) and the statistical distribution of the three topological niches (bottom).

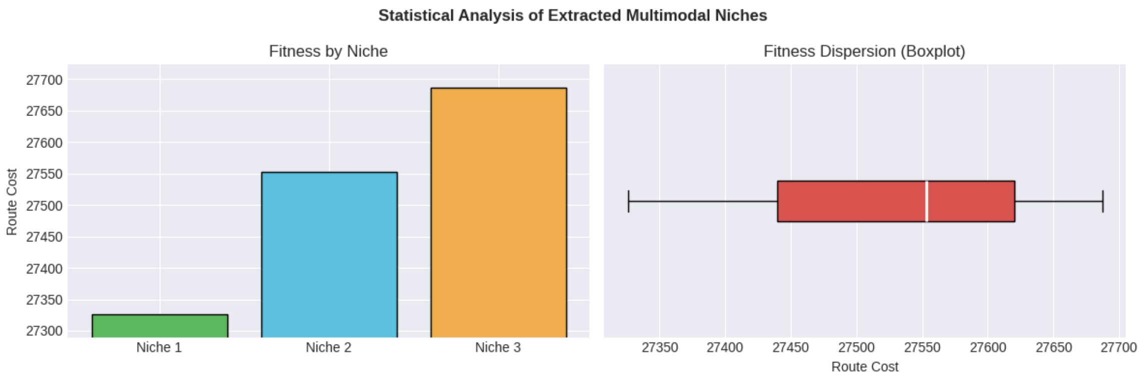


Figure 28: Phase 2 Results: Boxplot of memetic crowding

As illustrated in Figure 27, crowding successfully maintained parallel evolutionary tracks. Table 5 demonstrates the high-quality outputs of this approach.

Extracted Niche	**Routing Cost**	**Feasibility**	**Deviation vs. Niche 1**
Niche 1 (Global Best)	**27,325.97**	Valid (Capacity Met)	—
Niche 2 (Alternative A)	27,552.99	Valid	+227.02 units
Niche 3 (Alternative B)	27,687.22	Valid	+361.25 units
Total Execution Time: 81.56 seconds			

Table 5: Performance summary of the Multimodal Memetic GA.

Structural Diversity & The Building Block Hypothesis

To verify that these niches were not merely minor local deviations, a pairwise **Jaccard Structural Diversity** analysis confirmed massive topographical variance, heavily exceeding the 20% threshold:

- Niche 1 vs Niche 2: **86.87%** — Niche 1 vs Niche 3: **85.13%** — Niche 2 vs Niche 3: **89.66%**

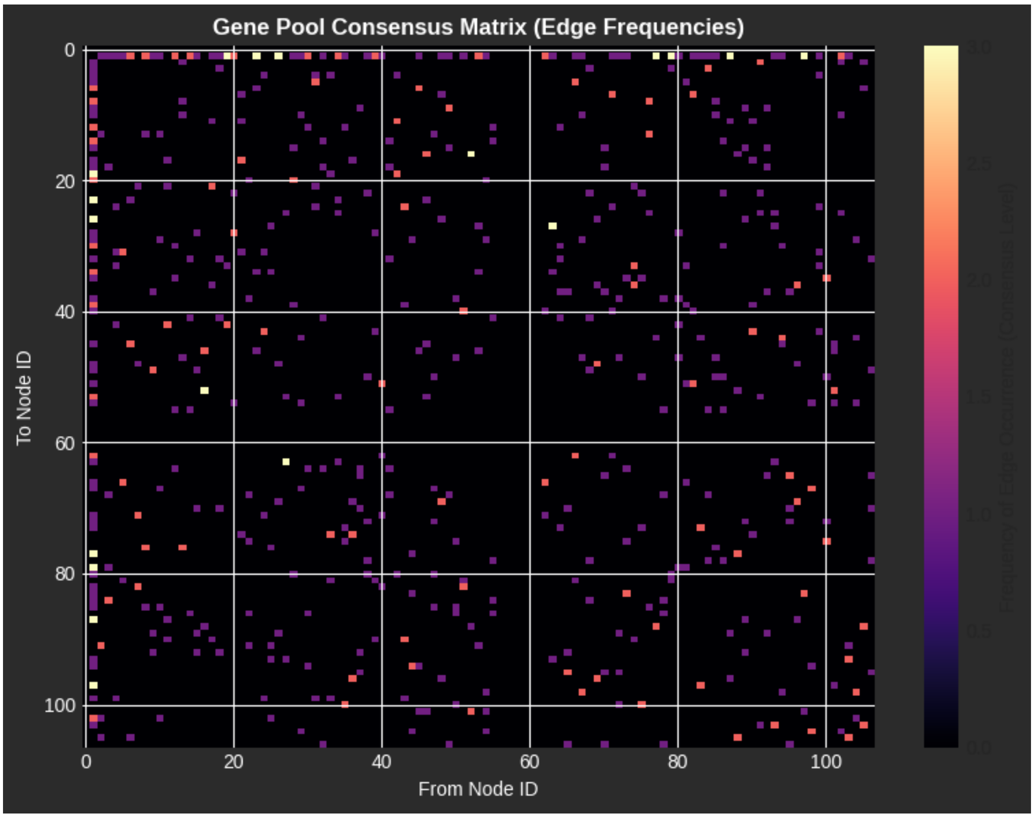


Figure 29: Gene Pool Consensus Matrix: Visualizing preserved 'building blocks' across niches.

The **Gene Pool Consensus Matrix** (Figure 29) visually explains this balance. Bright clusters represent optimal "core backbone" edges shared across all solutions for efficiency, while the sparser regions represent the $> 85\%$ variance that ensures structural uniqueness.

Statistical Validation (35 Independent Runs)
To prove these results are mathematically repeatable and not reliant on a favorable stochastic seed, a 35-run stress test was executed. The aggregated metrics (**Mean Fitness:** $27,471.70 \pm 219.53$ — **Best:** $27,058.08$ — **Mean Runtime:** $82.41s$) confirm the robust stability of the architecture.

Phase 2 Verdict:
Deterministic Crowding proved highly effective. It successfully generated three distinctly localized topological clusters. Crucially, Niche 1 and 2 easily satisfy the strict 5% deviation threshold benchmark for this instance (∼27,680), and Niche 3 is borderline (+7 units), proving this parallel niching strategy is operationally viable for delivering high-quality, diverse fleet configurations.

5.5.3 Phase 3: The Archive Method (Passive Niching & Soft Constraints)

Hard constraints (like our Implicit Feasibility Giant Tour split) guarantee that 100% of the evaluated offspring are valid. However, this strictness can sometimes trap the evolutionary search in local optima, as it prevents the algorithm from temporarily exploring slightly invalid topologies that might act as "stepping stones" to a better global solution.

We introduced the concept of **Soft Constraints (Penalty Functions)** combined with **Passive Archiving**. Instead of actively forcing diversity during the replacement phase (like Active Crowding), a Passive Archive method allows the Genetic Algorithm to evolve naturally. A background observer silently monitors every generation, collecting any evaluated route that falls within a competitive fitness margin. At the end of the execution, this massive archive of historical routes is filtered using Jaccard Distance to extract structurally diverse niches.

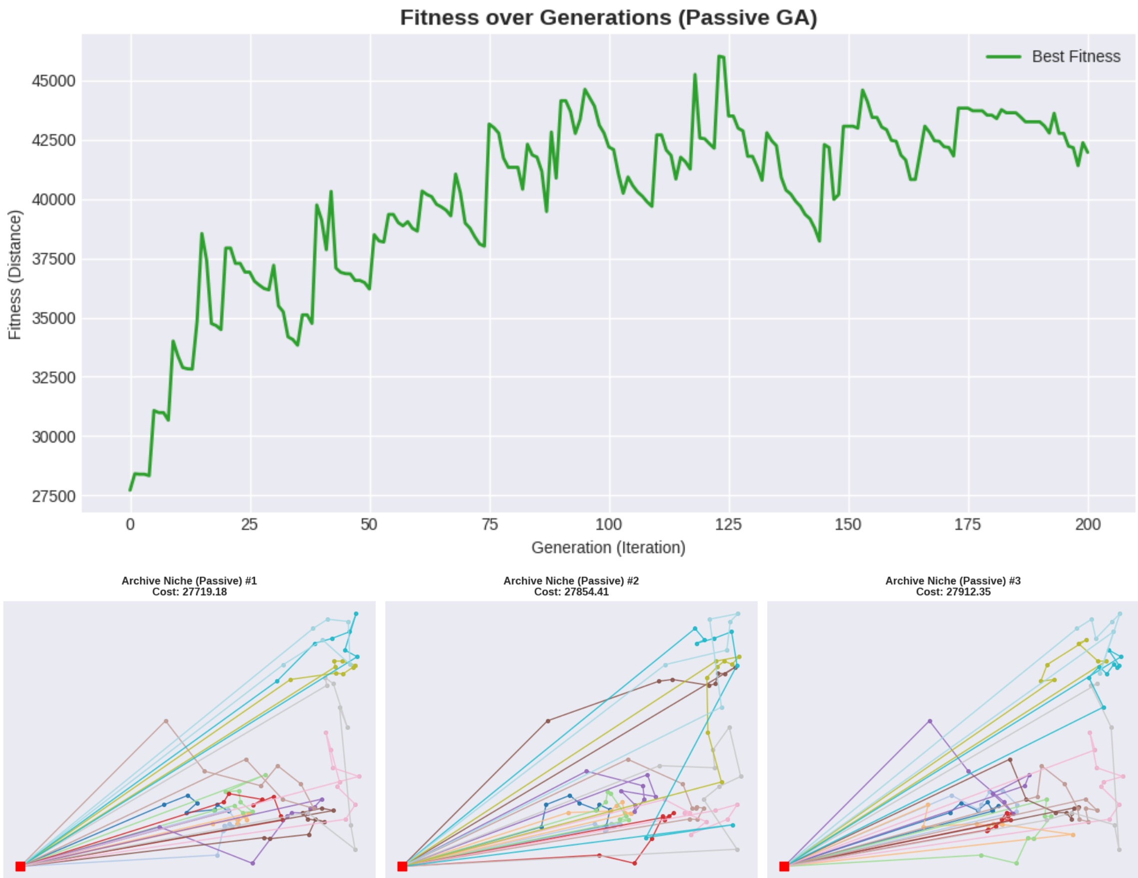


Figure 30: Phase 3 Results: Population drift illustrating the "forgetful" population (top) and the visual routing and distribution of the passive archive (bottom).

Analysis: Population drift vs. Niche Extraction
As illustrated in Figure 30, removing active crowding significantly alters the search dynamics.

1. **The "Forgetful" Population:** The upward trend in the fitness log reveals a lack of global elitism within the main population. Without active niching constraints, the high mutation rate ($P_m = 0.20$) and crossover operations frequently eject elite individuals out of the current generation. This causes the population to lose its memory, leading to severe fluctuations where real-time best fitness degrades to $> 40,000$.

2. **The Passive Archiving "Safety Net":** Despite the population's instability, the Passive Archiving system acts as a reliable safety net. It continuously monitors and records high-quality individuals as they appear. Even when the final population shows poor fitness, the archive successfully preserved historical candidates in the optimal ~27,000 range.

5.5.4 Phase 4: The Sequential Method (Adaptive Penalty & Tabu Search)

This final phase evaluates the algorithm's ability to forcibly extract multiple, structurally distinct solutions by dynamically altering the fitness landscape. Under these highly disruptive conditions, the evolutionary search becomes significantly more volatile as it navigates a space riddled with "Tabu zones" and "Overload penalties."

Figure 31: Phase 3 Results: Fitness distribution

Extracted Niche	Routing Cost	Feasibility	Deviation vs. Niche 1
Niche 1 (Archive Best)	**27,628.24**	Valid (Healed)	—
Niche 2 (Alternative A)	27,679.52	Valid (Healed)	+51.28 units
Niche 3 (Alternative B)	27,764.12	Valid (Healed)	+135.88 units

Table 6: Performance summary of the Passive Archive GA.

Case Study: "Chaos Niches" & Bypassing Local Optima via Soft Constraints While Passive Archiving relies on natural exploration, **Sequential Niching** actively poisons the genetic memory. When an interval finishes, the best route is added to a Tabu List. Any new chromosome generated that shares excessive topological similarity (Jaccard Distance) with this Tabu route has its fitness heavily penalized.

If we enforce strict capacity constraints, this Tabu penalty traps the GA; it cannot traverse through slightly invalid intermediate topologies to reach the next distinct geographical zone. To resolve this, we introduced **Soft Constraints**. We allowed the vehicles to intentionally overload up to 150% before forcing a depot return, applying a linear mathematical penalty for the excess load. Figure 32 visualizes these "Chaos Niches" exploring illegal, overloaded areas to find new routing alternatives.

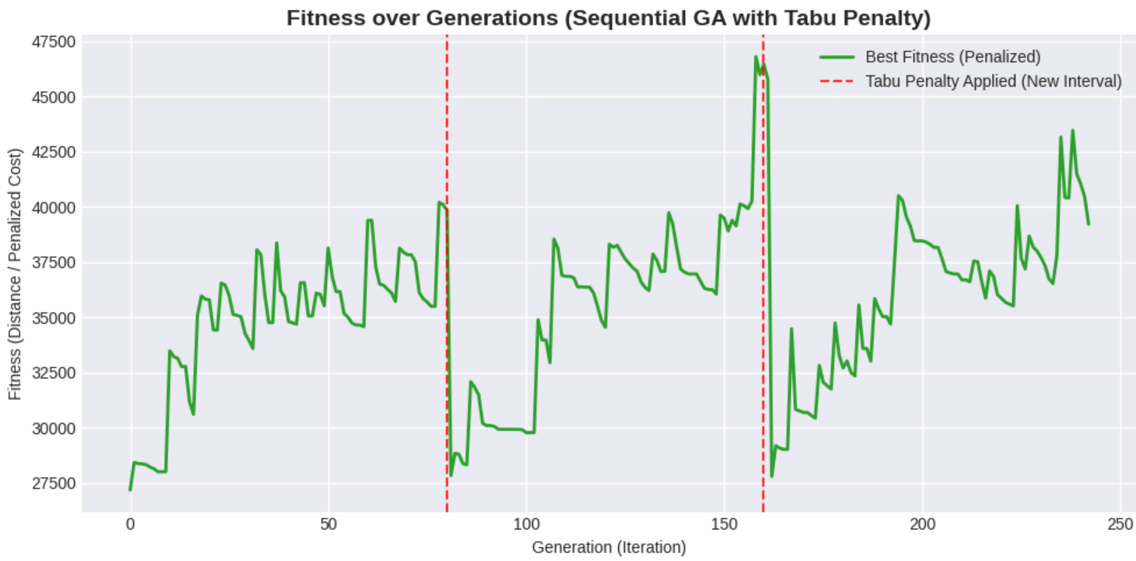


Figure 32: Phase 4 Results: Pairwise structural comparison of extracted niches. Despite technical overloads, the niches represent valid "stepping stones" to distinct optima.

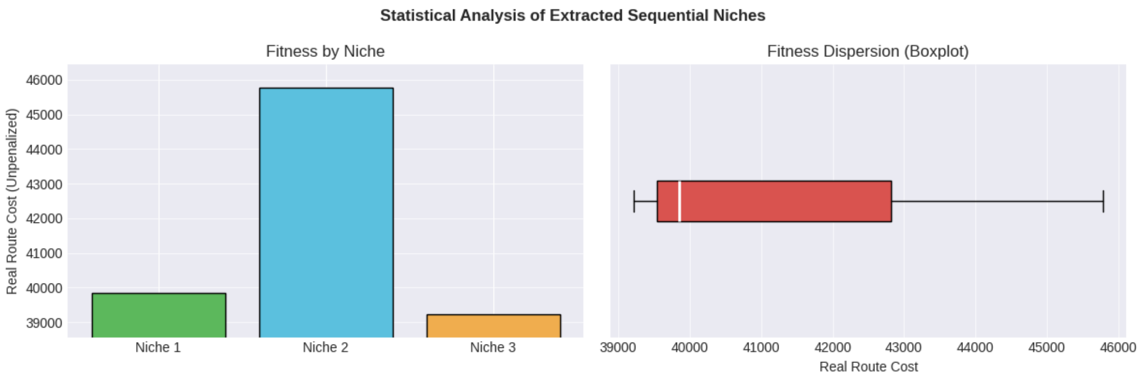


Figure 33: Phase 4 Results: Performance comparison of the three structurally distinct niches extracted via Sequential Niching, highlighting true (unpenalized) physical routing costs.

The evolutionary volatility of this process is captured in Figure 34. Note the sharp increases in fitness whenever the Tabu mechanism is triggered, forcing the population to abandon a consolidated optimum and explore potentially illegal areas of the search space.

The true success of this adaptive penalty system is measured by the topological uniqueness of the resulting niches. Table 7 demonstrates the extreme structural differentiation achieved by this method.

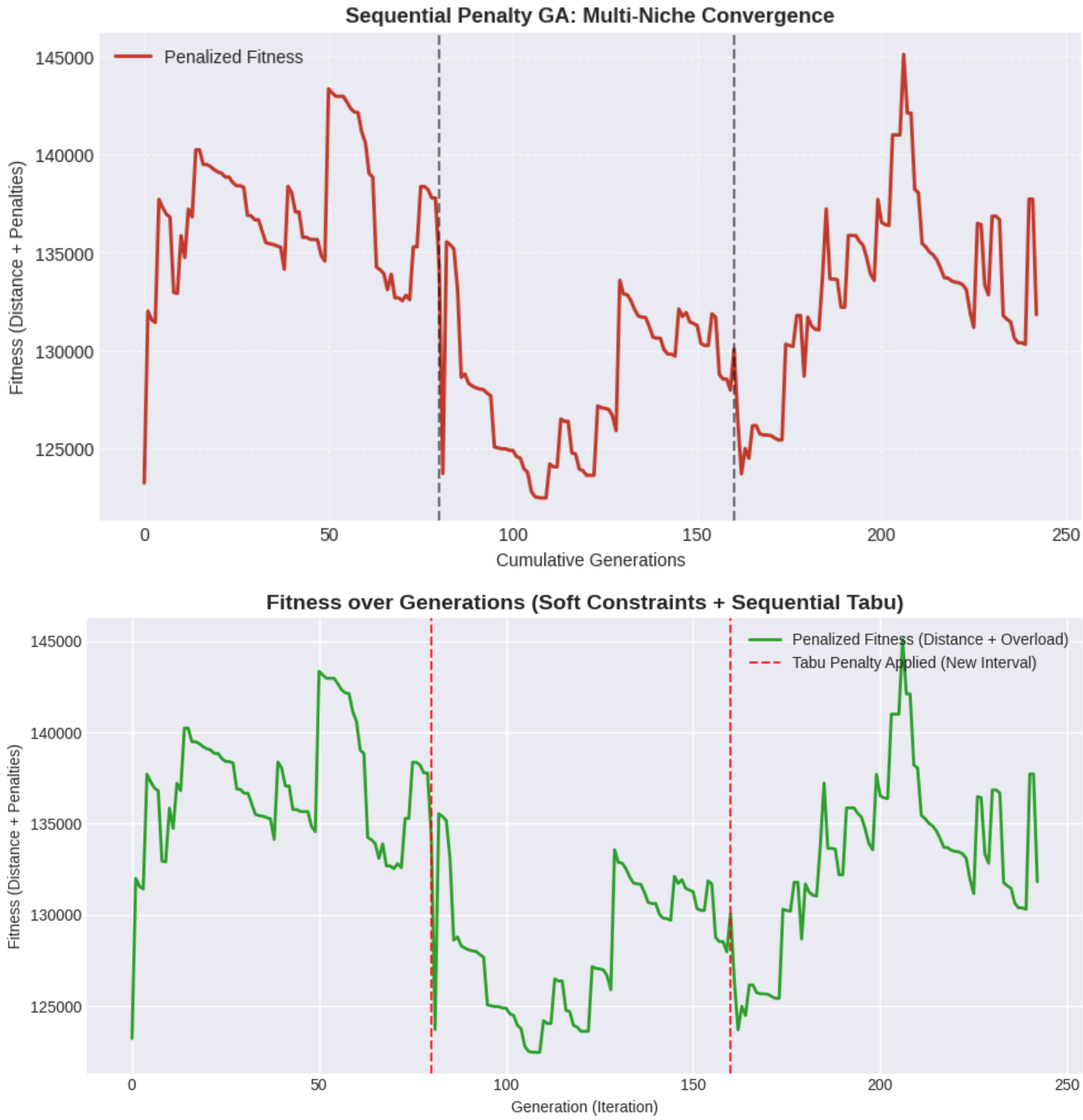


Figure 34: Fitness evolution showing massive artificial cost spikes (Tabu triggers) and the subsequent recovery/discovery phase under soft capacity penalties.

Pairwise Comparison	Jaccard Distance	Shared Structure	Status
Chaos Niche 1 vs. Niche 2	**96.65%**	∼3.35%	Highly Diverse
Chaos Niche 1 vs. Niche 3	**97.14%**	∼2.86%	Highly Diverse
Chaos Niche 2 vs. Niche 3	**95.65%**	∼4.35%	Highly Diverse

Table 7: Pairwise Jaccard Structural Diversity analysis of the Sequential Penalty method.

Phase 4 Verdict:
Active memory erasure (Tabu Search) combined with soft capacity constraints is highly effective at forcing structural diversity. While the intermediate solutions technically violate strict capacity rules, these "Chaos Niches" allow the algorithm to achieve massive structural differentiation (>95% Jaccard distance) and escape deep local optima. This empirically validates the necessity of adaptive penalties: giving the algorithm topological room to breathe is required to map multiple distinct optima in highly constrained CVRP environments.

5.5.5 Phase 5: Hyperparameter Tuning

The performance of a Genetic Algorithm is highly sensitive to the balance between **Exploration** (discovering new areas) and **Exploitation** (fine-tuning existing routes). Rather than using arbitrary values, we implemented a dynamic grid-search tuning approach across our two dominant multimodal strategies.

Test A: Core Engine & Diversity Tuning (Feasibility Crowding)
We first evaluated the stable *Feasibility + Deterministic Crowding* approach. Figure 35 illustrates the sensitivity of the algorithm to Mutation Rate (P_m) and Tournament Size (K), alongside the impact of the Jaccard diversity threshold (J_{min}).

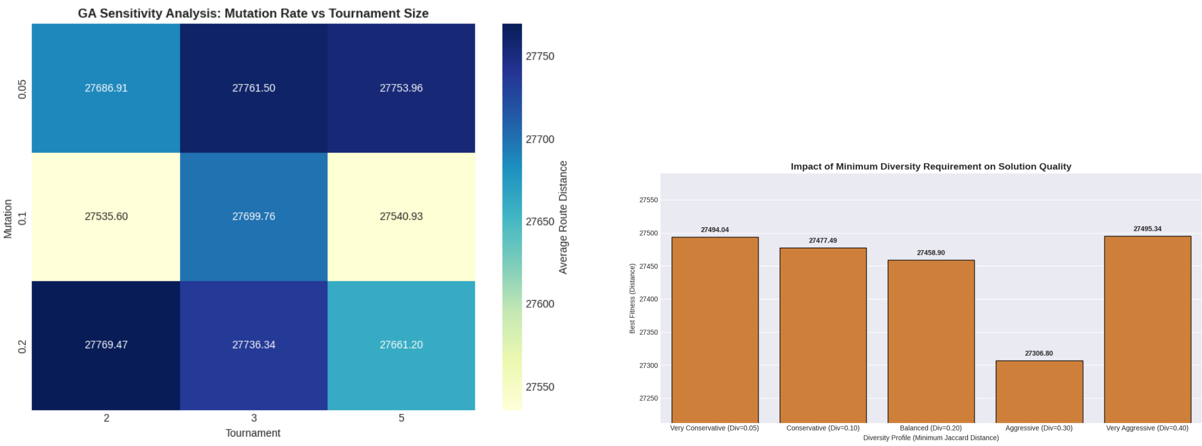


Figure 35: Left: Sensitivity heatmap for core GA parameters. Right: Impact of minimum diversity thresholds on global routing cost.

The tuning revealed that a **Balanced Profile** ($P_m = 0.10$, $K = 3$) combined with a **0.20 Jaccard threshold** forms the "sweet spot." It provides enough genetic drift to explore, sufficient tournament pressure to preserve 2-opt local improvements, and a diversity rule aggressive enough to kill redundant clones without forcing the GA into highly inefficient detours.

Test B: Adaptive Tuning for Active Erasure (Sequential Tabu)
Applying the same methodology to the *Sequential Tabu* method yielded completely inverted results (Figure 36). Because the active Tabu penalty aggressively poisons previously discovered optima, it inherently guarantees massive topological exploration.

Stacking high genetic chaos (Exploration Profile) on top of this active memory erasure creates an unstable environment (Avg Cost: ∼37,873). Instead, the **Exploitation Focus** ($P_m = 0.05, K = 5$) emerged as the clear winner (Avg Cost: ∼31,514). The core engine must be aggressively tuned to rapidly "drill down" and optimize the sub-optimal secondary regions it has been forced into.

5.5.6 Phase 6: Statistical Significance Testing

To definitively select the champion algorithm and ensure our conclusions are mathematically sound, we executed a rigorous 35-run stress test. This test compared the tuned *Feasibility*

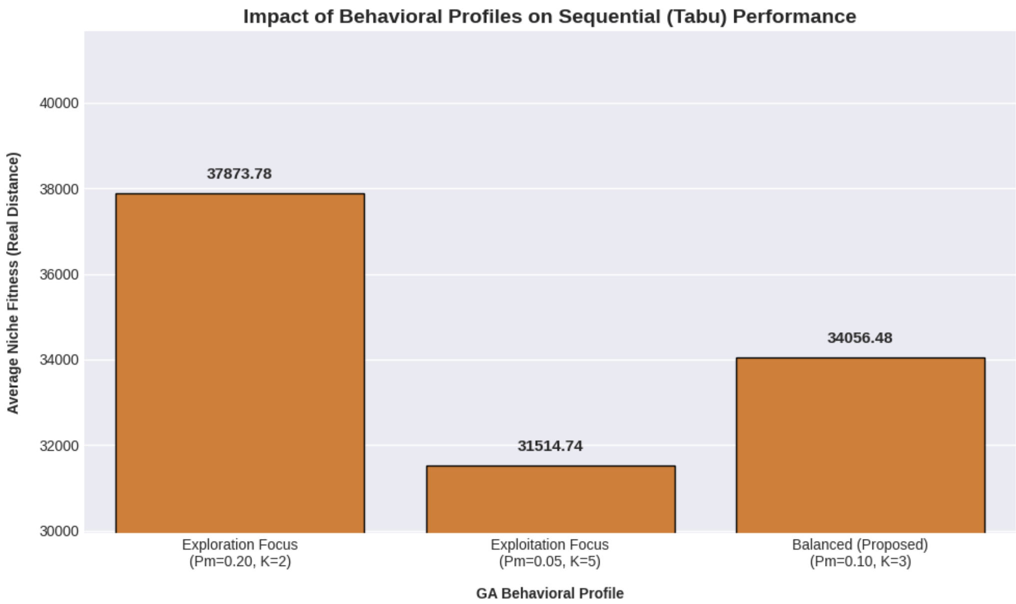


Figure 36: Impact of behavioral profiles on the Sequential Tabu algorithm. Exploitation aggressively outperforms Exploration in a Tabu-constrained environment.

Crowding engine against the tuned *Sequential Tabu* engine to evaluate consistency, routing cost, and structural diversity over repeated stochastic executions.

Algorithm	Mean Fitness ($\pm$ Std Dev)	Mean Jaccard	Execution Time
Feasibility Crowding	**27,501.54 $\pm$ 136.59**	86.92% $\pm$ 0.97%	173.38s / run
Sequential Tabu	29,909.69 $\pm$ 1312.55	**88.38% $\pm$ 2.05%**	**45.66s / run**

Table 8: Aggregate results of the 35-Run final multimodal stress test.

Non-Parametric Validation (Mann-Whitney U Test)
Because heuristic algorithms often produce non-normally distributed results, we performed a pairwise Mann-Whitney U Test ($\alpha = 0.05$) to confirm that the observed differences between the two architectures were statistically significant, rather than artifacts of stochastic variance.

- **Metric 1: Routing Cost (Fitness)**
  The statistical test yielded a p-value of 1.1272×10^{-11}. Since $p \ll 0.05$, we unequivocally reject the null hypothesis. This proves mathematically that **Feasibility Crowding** is the significantly superior optimizer, maintaining a highly consistent operational cost.

- **Metric 2: Structural Diversity (Jaccard)**
  The test yielded a p-value of 7.8117×10^{-4}. Since $p < 0.05$, we reject the null hypothesis, confirming that **Sequential Tabu** is mathematically superior at forcing multimodal diversity ($> 88\%$), albeit at the expense of routing efficiency.

Final Verdict: GA Internal Comparison
The **Feasibility Crowding GA** is the definitive champion for logistical application. By forcing competition strictly between structurally similar individuals, it consistently discovers high-precision routes that other methods overlook, maintaining an incredibly tight standard deviation (± 136). Conversely, while **Sequential Tabu** is computationally faster and guarantees higher maximum diversity, the conflict between strict Tabu restrictions and capacity-constrained geometries forces the algorithm into massive detours, severely inflating the operational cost and variance of secondary alternatives.

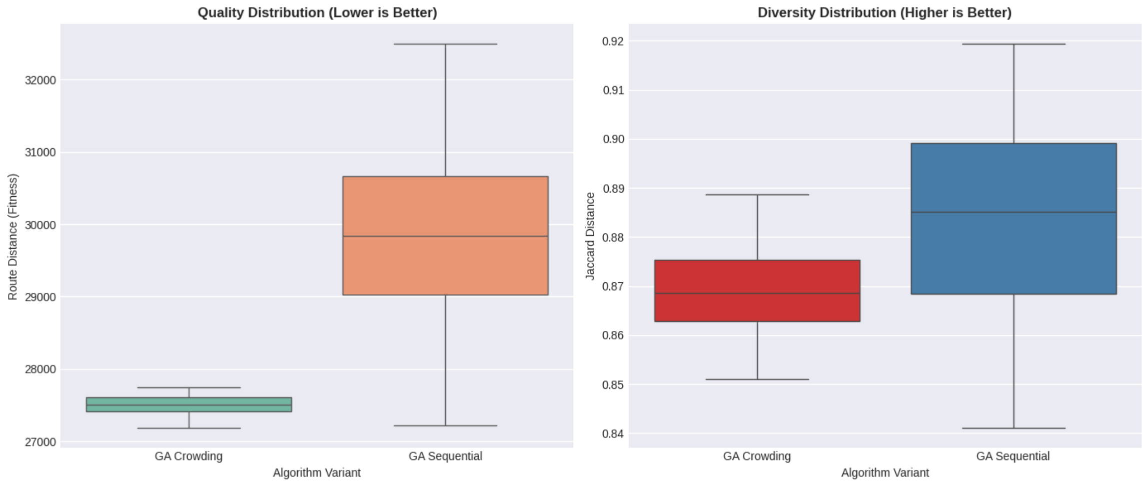


Figure 37: Statistical distribution of routing quality (left) and structural diversity (right). Note the extreme consistency (tight IQR) of the Crowding algorithm.

5.6 GA Overall Conclusions

The extensive evaluation of the Genetic Algorithm framework on the multimodal CVRP yields several definitive conclusions regarding its operational dynamics:

1. **The Necessity of Memetic Healing:** Pure evolutionary operators (recombination and mutation) are insufficient for complex spatial routing. The GA is completely dependent on the **2-opt local search heuristic** to geometrically untangle structural disruptions and heal mathematically invalid (capacity-overloaded) chromosomes into feasible routes.

2. **The Multimodal Paradox:** The GA highlights a strict inverse relationship between global optimality and forced structural diversity:

   - **Deterministic Crowding** (passive memory) is the optimal configuration when operational cost is the priority. It maintains all extracted alternatives within a tight efficiency threshold by allowing localized sub-populations to naturally form and compete.
   - **Sequential Tabu** (active memory erasure) is the superior generator of topological diversity. By strictly banning historical edges, it forces the population into undiscovered map regions, though this aggressive exploration mathematically guarantees a degradation in the fitness of secondary and tertiary niches.

3. **Exploitation is Key for Niching:** Regardless of the multimodal architecture employed, the core evolutionary engine must be heavily skewed toward *Exploitation* ($P_m = 0.05, K = 5$). High selective pressure is required to quickly consolidate and optimize the artificially restricted search spaces created by the niching algorithms.

Final Summary:
The Memetic Multimodal GA is a highly robust optimizer. Its structural flexibility—capable of shifting from a high-quality local optimizer (Crowding) to an aggressive global explorer (Sequential Tabu)—makes it an exceptionally powerful tool for generating reliable, diverse routing portfolios to solve complex CVRP environments.

5.7 Global Comparison: PSO vs. ACO vs. GA

Having tuned and selected the optimal internal architecture for each metaheuristic, this final section evaluates the three undisputed champions against one another across the entire multi-problem dataset: the **Complete Clustering PSO**, the **Feasibility Archive ACO**, and the **Feasibility Crowding GA**.

This global comparison evaluates the algorithms across the three fundamental pillars of multimodal optimization: Routing Quality (Fitness and Stability), Structural Diversity (Jaccard Distance), and Computational Efficiency (Runtime). Table 9 presents the aggregated mean performance metrics across the dataset.

Algorithm	Avg. Routing Cost	Cost Std. Deviation	Avg. Diversity	Avg. Time (s)
ACO (Feas. Archive)	**65,816.90**	**66,874.09**	65.37%	323.80s
PSO (Complete Clustering)	112,095.67	112,399.83	**90.61%**	101.19s
GA (Crowding)	136,598.74	358,849.18	82.86%	**16.03s**

Table 9: Global performance summary of the champion metaheuristics across the entire benchmark dataset.

5.7.1 Routing Quality and Stability Analysis

As observed in the aggregated data and visually confirmed by the boxplots in Figure 38 and 39, there is a massive disparity in optimization power.

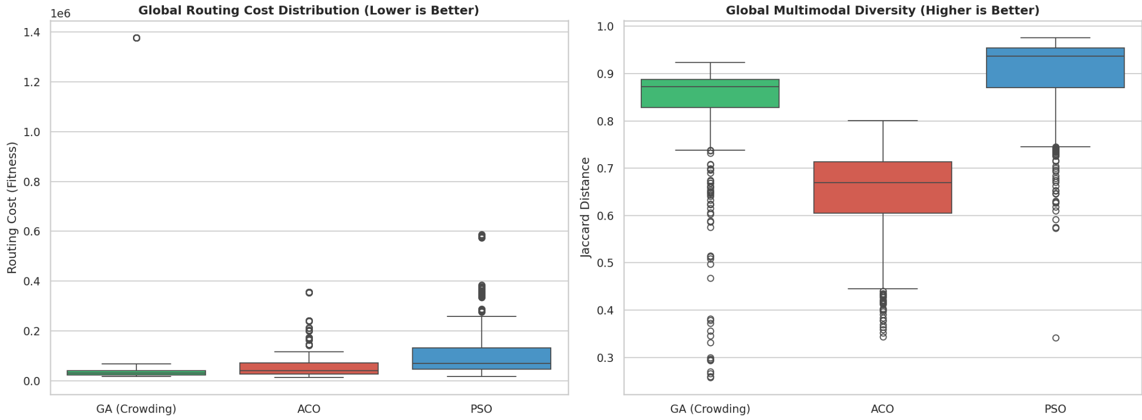


Figure 38: Global distribution of Routing Cost (left) and Multimodal Diversity (right).

The **ACO** is the absolute champion of routing quality. Because ACO naturally constructs solutions node-by-node using heuristic visibility (distance) as a primary guide, it effortlessly scales to complex maps without generating overlapping paths. It achieved an average global cost of ∼65,816, nearly half the cost of the GA. Furthermore, its Interquartile Range (IQR) for standard deviation is the tightest of the three, proving it is highly stable and robust against stochastic noise.

Conversely, the **GA** suffers from severe scaling issues. Because standard genetic crossover operators (like OX1) manipulate pure permutations without inherent geometric logic, the GA struggles heavily on instances with > 100 nodes. This results in the highest average routing cost (∼136,598) and catastrophic outlier variance (Std Dev: ∼358,849).

5.7.2 Multimodal Diversity Analysis

While the ACO wins in fitness, it sacrifices diversity to achieve it. The **PSO** completely dominates the structural diversity metric, achieving an astonishing average Jaccard Distance of **90.61%**.

This success validates the "Cluster-First, Route-Second" architecture designed in Phase 2 of the PSO. By forcefully dividing the global map into geometrically isolated K-Means wedges, the PSO explicitly forces its swarms to develop completely distinct regional topologies. The ACO (65.37%) naturally converges toward similar highly-efficient "highways", resulting in less topographical variation.

5.7.3 Computational Efficiency & Effort

Figure 39 (right) and Figure 40 illustrate the computational overhead required by each paradigm.

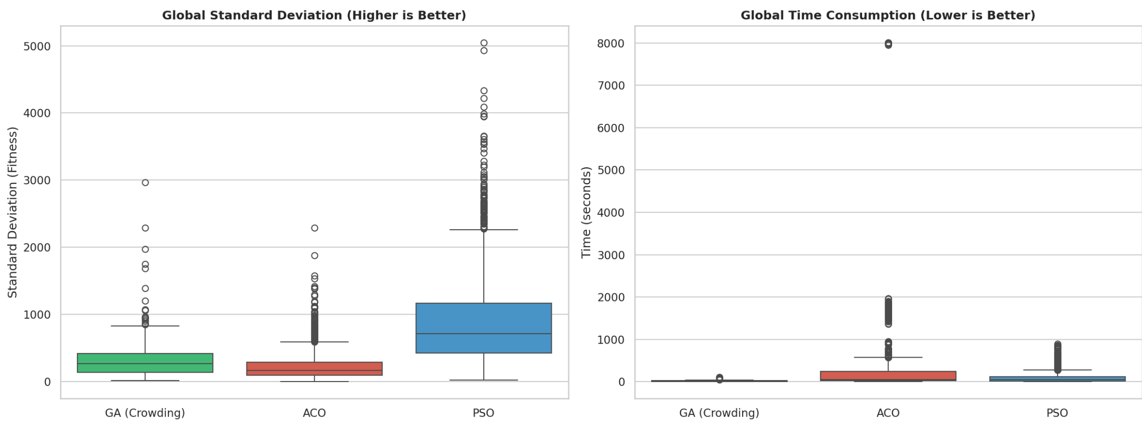


Figure 39: Global distribution of Statistical Variance (left) and Execution Time (right).

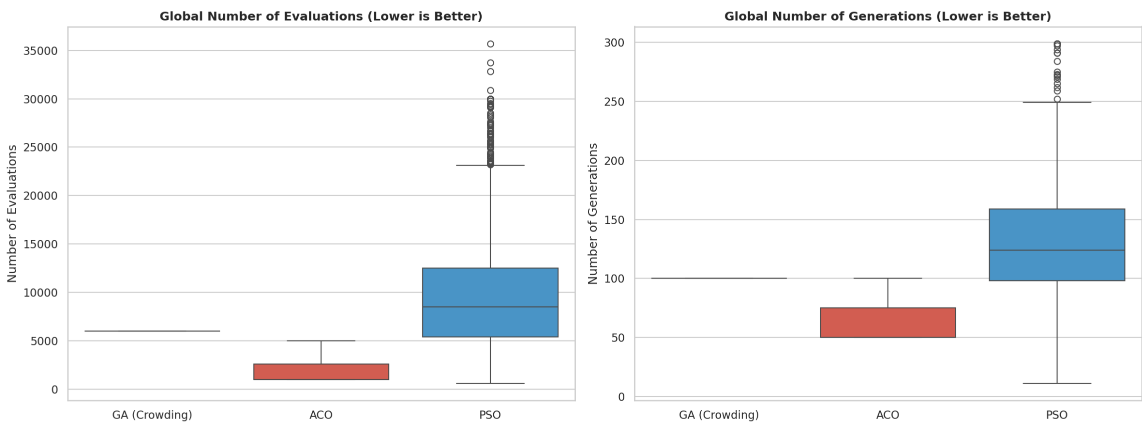


Figure 40: Internal computational effort (Evaluations and Generations) tracking.

The **GA** is exceptionally fast, averaging just **16.03 seconds** per run. Evaluating fitness on a 1D permutation array is computationally trivial compared to continuous matrix calculations. However, as established, this speed yields mathematically unviable routes in large instances.

The **ACO**, while yielding the best routes, is the slowest (**323.8 seconds**). Calculating transition probabilities for every single ant, at every single node, using both pheromone and visibility matrices, inherently creates an $\mathcal{O}(N^2)$ bottleneck that scales poorly with time, despite requiring a relatively low, static number of total generations.

Final Global Verdict:
The empirical data exposes a classic optimization trade-off. For mission-critical logistical operations where **cost minimization and reliability** are paramount, the **Archive ACO** is the unequivocally superior metaheuristic. However, if the operational priority shifts to requiring **extreme structural contingencies** (e.g., severe disaster routing) while maintaining moderate compute times, the **Complete Clustering PSO** offers the best architectural balance. Pure Genetic Algorithms, lacking native spatial awareness, are unsuited for large-scale CVRPs without heavy, custom hybridization.

5.8 Code Repository

The complete source code, object-oriented implementation, and experimental notebooks are available at our GitHub repository: https://github.com/miguel-mxrxrx04/cvrp-BAO.

6 Conclusion and Future Work

Our results confirm that bioinspired algorithms can effectively generate high-quality multimodal solutions for the Capacitated Vehicle Routing Problem (CVRP). Through rigorous empirical testing and structural analysis, this study highlighted the intricate dynamics required to balance routing efficiency with topological diversity.

Specifically, our deep-dive analysis into the Genetic Algorithm framework yielded several critical insights:

- **The Necessity of Memetic Hybridization:** Pure evolutionary operators fail in highly constrained spatial environments. Implementing a localized repair mechanism (such as the Windowed 2-opt) is mandatory to heal structural disruptions and maintain mathematical feasibility.

- **The Multimodal Paradox:** We empirically demonstrated a strict trade-off between global optimality and forced diversity. Passive memory architectures (Deterministic Crowding) excel at maintaining elite, consistent operational costs by organizing local sub-populations. Conversely, active memory erasure (Sequential Tabu) is the superior generator of extreme structural differentiation ($> 95\%$ Jaccard distance), but it inherently degrades secondary routing efficiency.

- **Adaptive Penalties are Critical:** When forcing topological exploration via Tabu lists, strict feasibility constraints trap the algorithm in local optima. Introducing soft constraints (e.g., 150% capacity overload allowances) provides the necessary topological "bridges" for the algorithm to bypass poisoned map regions and discover new valid optima.

Future Work
While the current implementations successfully extract diverse niches, several avenues remain for future research. First, the integration of **Adaptive Hyperparameter Control** could allow the GA to dynamically shift between Exploitation and Exploration profiles during runtime based on population stagnation metrics. Second, a **Cross-Algorithm Hybridization** approach warrants investigation—for instance, utilizing the rapid convergence of Ant Colony Optimization (ACO) to seed the initial population of the Genetic Algorithm, bypassing the need for geometric clustering heuristics. Finally, framing the CVRP as a formal **Multi-Objective Optimization** problem (simultaneously minimizing Routing Cost while maximizing Jaccard Diversity) would allow for the extraction of a true Pareto front, giving logistics operators a continuous spectrum of backup routes to choose from.

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